CHAGEE

CHAGEE SINGAPORE

THE OPPORTUNI-TEA — UNLOCKING GROWTH AT CHAGEE SG WITH AI

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN · OCTOBER 2026



SPEAKER NOTES · TITLE

Good morning. We are Group 3. Our company is Chagee in Singapore, and our story has three solutions: online retailing, matching each online order with a convenient pickup; in-store management, serving that demand well at the counter; and new store expansion, opening where customers are underserved.

Each solution has its own decision-maker: the digital commerce manager with operations, the operations manager or outlet GM, and the expansion manager. After the close, an optional extension on Australia and an appendix hold the detail for questions.

Timing: title and agenda together, about twenty seconds.

AGENDA

Spilling the Tea

01 **Introduction**
Chagee Singapore, our scope and why

02 **Executive Summary and Business Plan**
Two key challenges and one strategic decision

03–13 **Our Business Recommendations**
Online retailing · in-store management · new store expansion

14 **Execution Plan**
Three pilots and when to proceed

\+ **Business Impact**
What the plan is worth, in our model · appendix, draft

15 **Q & A**

\+ **Appendix**
Earlier slides (data trail, scope, membership), derivations, tables and sources; the Australia extension comes just before it



SPEAKER NOTES · AGENDA

Not presented: go straight from the title to the introduction. For reference: the introduction and the executive summary; our recommendations as three solutions, each with its decision-maker (online retailing for the digital commerce manager with operations, in-store management for the operations manager or outlet GM, new store expansion for the expansion manager); the execution plan, which is our recommendation slide with the pilots and when to proceed; and Q&A. Business impact has no main slide yet: the line opens the draft in the appendix. The Australia extension sits after Q&A, just before the appendix, which now also holds the data trail, the earlier scope slide and the membership chapter. Click any line to jump there.

Today, we're presenting Chagee

A tea brand most of you know: order in the app, collect at the counter, or have it delivered to your location.



Our scope

Chagee Singapore's management and its customers

Outlets are **company-owned, not franchised**, so the counter, roster and equipment are Chagee's own calls.

Why Chagee?

1. **Many stores, managed by one company**, one data system, with lots of data appendix →
2. **App-centric**: collects personal and transactional data
3. **Fast-growing, playful**: rapid expansion, room for improvements with AI
4. **An insider's view**: we were fortunate to interview a staff member (anonymous)

FROM FARM TO CUP · OUR PROJECT FOCUS IS DOWNSTREAM

Source
Yunnan tea; local milk and fruit

Warehouse
Restocking from forecasts

Upstream data and decisions sit with **Chagee corporate**

Outlets
Make and hand over

via Chagee App

Last mile
Collection or delivery

Downstream data is analysed and decisions are made by **Chagee SG** (APAC HQ)

OUR FOCUS | The downstream operational challenges: some of the most pressing issues for Chagee Singapore's management.

SPEAKER NOTES · CHAGEE SINGAPORE AND OUR SCOPE

Today we are presenting Chagee: a tea brand most of you know. Order in the app and collect at the counter, or have it delivered; the app is for ordering ahead, and delivery goes through third-party apps such as Grab (anonymous Chagee source).

Our scope is Chagee Singapore, its management and customers. Singapore outlets are company-owned, not franchised, as our project owner confirmed, so the counter, roster and equipment are Chagee's own decisions. If asked: group-wide, about 88% of FY 2025 net revenue came from franchising (secondary source); that is global context, not Singapore.

Why Chagee: many stores under one company and one data system (anonymous Chagee source); app-centric: an order-ahead, self-pickup app (App Store listing), in-app ordering after the reopening queues (Vulcan Post), and personal and transactional data collected (Form 20-F, FY 2025); fast-growing and playful, with room for AI. The app's share of Singapore orders, and whether searches or location are logged, are to confirm. And we interviewed a Chagee staff member, who asked not to be named.

Farm to cup: tea comes from Yunnan through standard channels, only fresh milk and fruit are local (YF Chuhai), and restocking is planned from forecasts (Form 20-F). Downstream, the outlet makes each cup and the app links it to collection or delivery; which of these Chagee records is to confirm. Upstream sits with Chagee corporate; downstream is analysed and decided by Chagee Singapore, so that is our focus. Caution: this split, and the APAC HQ label, come from our team's storyboard with no public source; confirm before presenting.

So we focus on the downstream operational challenges. That leads into the executive summary. About one minute.

Our business plan to solve Chagee's problem statements



CASE STUDY · OUR INTERVIEW

"Mei"

Membership Operations · APAC, Chagee

Persona: name changed, interviewee anonymised

WHAT MEI TOLD US

Through the interview, we learnt about Chagee's **business model**, its **local operations**, and some of its most **pressing concerns and strategic needs**, which AI could potentially help solve.

TWO KEY CHALLENGES + ONE STRATEGIC DECISION POINT

KEY CHALLENGE

Online retailing

Use data for better sales, customer loyalty and cost management. Customer management and reduction of wastage.

PROPOSAL 1 · OUR TEAM

Capacity-aware ordering, targeted discounts and rewards for convenient self-pickup

slides 03 to 07 →

DECISION-MAKER

Digital commerce manager, with CRM and operations

KEY CHALLENGE

In-store management

A faster, smoother counter: in-store operations management for shorter waits and a crew matched to demand.

PROPOSAL 2 · OUR TEAM

Forecast staffing requirements and compare practical ways to organise people and machines

slides 08 to 11 →

DECISION-MAKER

Operations manager / outlet GM

STRATEGIC EXPANSION DECISION

New store openings

Where to open next, so a new store is profitable and adds customers instead of moving them.

PROPOSAL 3 · OUR TEAM

Identify underserved search clusters, evaluate candidate locations and assess net additional demand and financial outcomes

slides 12 and 13 →

DECISION-MAKER

Expansion manager

SPEAKER NOTES · EXECUTIVE SUMMARY

Our business plan starts from Chagee's own problem statements. We interviewed a member of Chagee's Membership Operations team for APAC, shown here as a persona, "Mei": the name is changed and the interviewee asked not to be cited openly.

Through the interview, we learnt about Chagee's business model, its local operations, and some of its most pressing concerns and strategic needs, which AI could potentially help solve.

We frame them as two key challenges and one strategic decision point. Challenge one, online retailing: using data for better sales, customer loyalty and cost management, through customer management and reducing wastage. Challenge two, in-store management: a faster, smoother counter, through in-store operations management for shorter waits and a crew matched to demand. And the strategic decision: new store openings, where to open next so a new store is profitable and adds customers rather than moving them from nearby outlets.

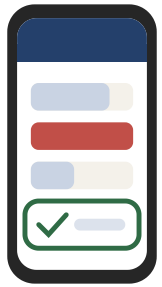
The top of each column is what we heard; the shaded part below is our team's proposal, not something Chagee asked for, with its decision-maker in green. Solution one, online retailing: help customers choose suitable outlets and collection times using demand and readiness predictions, for the digital commerce manager with operations. Solution two, in-store management: forecast staffing requirements and compare practical ways to organise people and machines, for the operations manager or outlet GM. Solution three, new store expansion: identify underserved search clusters, evaluate candidate locations and assess net additional demand and financial outcomes, for the expansion manager. We do not address wastage in this deck; the membership-squad analysis is in the appendix. We walk through them in this order next. About forty-five seconds.

DECISION-MAKER **Digital commerce manager, with CRM and operations**

Course link · Topic 1, online retailing

Coordinate online demand, rewards and pickup to improve customer experience and contribution

BUSINESS PROBLEM Peak congestion can discourage purchases, promotions can subsidise orders that would happen anyway, and fulfilment channels have different costs.



1

Capacity-aware ordering

Offer only pickup times and nearby outlets that have room, window by window.

Predicted demand and preparation speed · **calculated** capacity from staff, equipment and queued work

slide 04 →



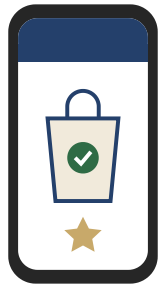
2

Targeted discounts

Discounts only, never surcharges, timed and targeted where an offer is likely to add an order.

A model estimates demand and each member's incremental response · **policy** sets the offer

slide 05 →



3

Reward convenient self-pickup

Show transparent savings and an eligible reward when a customer collects it themselves.

Two journeys, valued apart · delivery to pickup, and third-party pickup to Chagee-app pickup

slide 06 →

ONE JOURNEY | A feasible pickup, an offer only where it adds an order, and a reward for collecting: the business case and the decision are on slide 07.

SPEAKER NOTES · ONLINE RETAILING: THREE CONNECTED INITIATIVES

Solution one, online retailing, for the digital commerce manager, supported by CRM and operations. The business problem has three parts: peak congestion can discourage purchases; promotions can subsidise orders that would have happened anyway; and the fulfilment channels, Chagee-app pickup, in-app delivery and Grab, carry different costs.

Click through three connected initiatives in one app journey. One, capacity-aware ordering: offer only pickup times and outlets with room. Two, targeted discounts: discounts only, timed and targeted where an offer is likely to add an order. Three, reward convenient self-pickup: transparent savings and an eligible reward, with two different journeys valued separately.

Where the AI is: predictions of demand, preparation speed and each member's incremental response to an offer. Capacity is calculated, and the offer amounts are policy. The next three slides unpack one initiative each; slide 07 has the business case and the decision. About forty seconds.

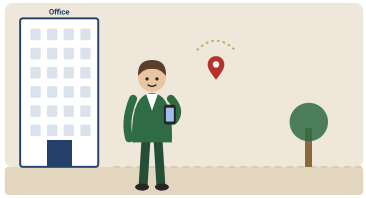
DECISION-MAKER **Digital commerce manager, with CRM and operations**

Match each online order with a convenient pickup option

INITIATIVE ONE Forecast demand and estimate available capacity in ten-minute windows, then offer feasible pickup times and nearby outlets. Switching stays optional.

1 CUSTOMER

I want to order ahead.



I want to order ahead and know when it will be ready.

OFFICE WORKER, LUNCHTIME

The customer's goal: convenient collection with a dependable time estimate.

2 INTELLIGENCE

Match the order to capacity.

Current orders and drink mix
INPUT · RECORDED TODAY

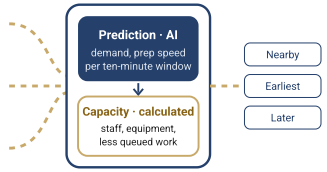
Staff, equipment and queued work
CAPACITY · CALCULATED

Forecast incoming demand
PREDICTION · AI

Preparation speed
PREDICTION · AI

Customer travel time
INPUT · NEEDS LOCATION, WITH CONSENT

Preferred collection time
INPUT · THE CUSTOMER ENTERS IT



AI predicts demand and preparation speed. **Capacity is calculated** from staff, equipment and outstanding work; the app offers only outlet and time options with room.

Illustrative demonstration: made-up outlets and values. Predictions are estimates; capacity is a calculation; only windows with room are offered.

3 APP EXPERIENCE

Choose your pickup plan.

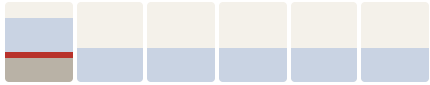
Nearby a convenient outlet and its estimated ready time.

Earliest collection the earliest feasible option, including travel.

Later pickup a collection window that suits the customer.

THE CHOSEN OUTLET'S WORKLOAD, NEXT HOUR

MRT-station outlet
walk 9 min · 14 cups a window



The customer chooses; the order joins that outlet's workload. A confirmed order is never moved automatically.

DECISION-MAKER **Digital commerce manager, with CRM and operations**

Match each online order with a convenient pickup option

INITIATIVE ONE Forecast demand and estimate available capacity in ten-minute windows, then offer feasible pickup times and nearby outlets. Switching stays optional.

PICKUP PLANNER · ILLUSTRATIVE DEMO · MADE-UP OUTLETS AND WORKLOADS



Pickup plan

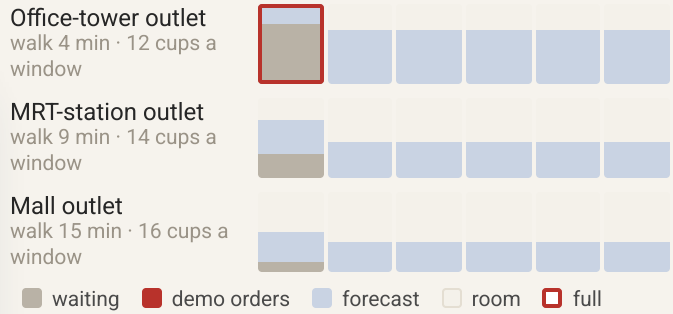
Jasmine milk tea · 1 cup · now 12:40

Nearby 13:00
Office-tower outlet · walk 4 min
drink ready 13:00

Earliest collection 12:50
MRT-station outlet · walk 9 min
drink ready 12:50

Later pickup 13:10
Office-tower outlet · walk 4 min
collect 13:10–13:20

OUTLET WORKLOAD · NEXT HOUR, CUPS PER TEN-MINUTE WINDOW



Options are optional and only use windows with room; a confirmed order updates capacity and never moves.

Value. Customer: clearer choices, convenient collection, more reliable pickup estimates. Chagee: online demand coordinated with capacity, spare capacity used, orders potentially retained. No improvement is claimed; a pilot would measure it.

NEXT Initiative two: offer a discount only where it is likely to add an order.

SPEAKER NOTES · MATCH EACH ONLINE ORDER WITH A PICKUP OPTION

Initiative one, capacity-aware ordering. The customer orders ahead; the app should offer only pickup times and outlets that can actually deliver. Two kinds of number feed it, and we keep them apart. Predicted, by AI: incoming orders and preparation speed for each ten-minute window. Calculated, not predicted: each window's capacity, from the people on shift and the working equipment, less the cups already queued.

Step one, the customer: an office worker wants to order ahead with a time they can trust. Step two, the intelligence: queued orders, staff and equipment, forecast demand, preparation speed, travel time and the customer's preferred time. Dashed inputs would need data access: rosters and machine status, and location with consent. Step three, the app: three choices, nearby, earliest collection including travel, and a later window.

Then the demo, all illustrative: made-up outlets, workloads and walking times. Preparation runs while the customer walks, so collection is whichever comes later, the walk or the drink. Move the demand slider: the nearby outlet fills and the earliest option moves elsewhere. Confirm an order: capacity updates at once, and the order stays in its window even if demand changes. Switching outlet or time is always the customer's choice. We claim no measured improvement here; the business case is on slide 07.

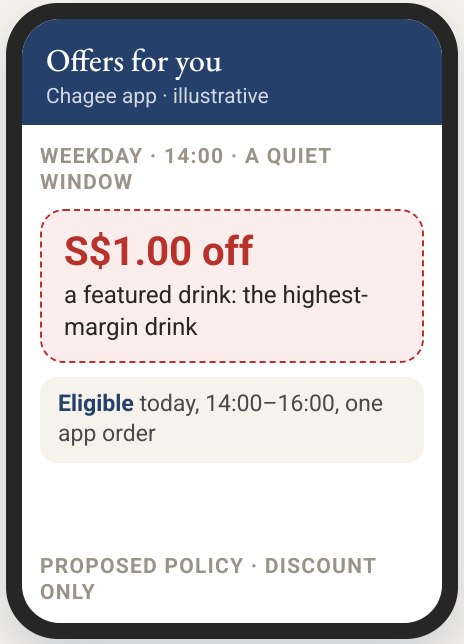
Course link, Topic 1, online retailing: integrating online and offline channels. Research supports the direction, not the size: in one restaurant, longer waits went with customers leaving and returning less often (De Vries and co-authors, 2018), and a queueing model finds pooled capacity serves app and platform orders more profitably (Wang and co-authors, 2024). Neither is a Chagee estimate. About one minute.

DECISION-MAKER **Digital commerce manager, with CRM and operations**

Offer a discount only where it is likely to add an order

INITIATIVE TWO Targeted promotional pricing: discounts only, never surcharges. The offers below are the team brief's proposed policy, not proven optimal decisions.

THE APP, BY SCENARIO · ILLUSTRATIVE



Printed: the off-peak scenario; all four are in the table.

Why this offer, in this scenario

Spare capacity in a quiet window. An offer pays only if it brings orders that would not otherwise happen; a model estimates who that is likely to be.

Proposed policy · all four scenarios

PROPOSED, NOT PROVEN

SCENARIO	OFFER	WHEN
Off-peak	up to S$1.00, highest-margin drink	quiet windows
Peak	none; S$0.50 off a pre-order collected later	busiest hours
Weather or event	S$0.50 hot; S$1.00 normal or rain; none at events	by forecast
Inactive member	up to S$1.00 off a favourite; then a free upsize	7 days, then 14 more

Where AI comes in

A fixed scenario rule is not AI. A model would estimate **demand in each window** and each member's **incremental response** to an offer; the offer then goes where expected incremental contribution is highest, within the member's preferences and the outlet's capacity.

BREAK-EVEN | At S$1.00 off and about S$3.54 contribution a cup, an offer pays only if at least 28% of redeemed orders are new. Test it against a no-offer group.

NEXT Initiative three: reward customers who collect it themselves.

SPEAKER NOTES · TARGETED DISCOUNTS

Initiative two, targeted discounts: targeted promotional pricing, so every adjustment is a discount, never a surcharge. Click through the four scenarios on the phone. Each shows the offer as it would appear in the app, its eligibility window, and the business reason.

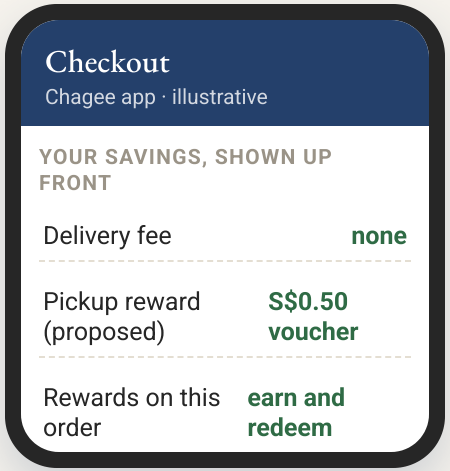
Off-peak: up to S$1.00 off the highest-margin drink, in a quiet window, to fill spare capacity. Peak: no vouchers; S$0.50 off a pre-order only if collection moves into a window with room, because ordering earlier alone does not cut the peak's production workload; and limiting Grab delivery when slots are full, which depends on what Grab's merchant tools allow. Weather and events: S$0.50 off in hot weather, S$1.00 in normal weather or rain, none during nearby events. A member becoming inactive: up to S$1.00 off their favourite for 7 days, then a free upsize after 14 more days.

These amounts and timings are the team brief's proposal, not optimal decisions. A fixed rule by scenario is not AI. The AI part is the model behind it: it estimates demand in each window and each member's incremental response, so an offer goes only where it is likely to add an order, within the member's preferences and capacity. The break-even is simple: a discount is paid on every redeemed order, including those that would have happened anyway, so with about S$3.54 contribution a cup, at least 28% of redeemed orders must be new. Only a no-offer comparison group can measure that. For context, delivery platforms with dynamic pricing show higher demand than static ones in China (Tong and co-authors, 2020); that is an association elsewhere, not a Chagee estimate. About one minute.

Reward collecting it yourself, and count only the cost Chagee avoids

INITIATIVE THREE Show transparent savings and an eligible reward at checkout. Two different journeys, valued separately; moved orders shift revenue, they do not add it.

THE CUSTOMER'S CHECKOUT · ILLUSTRATIVE



Printed: delivery → chagee-app pickup, the savings step; the other journeys are in the notes.

Delivery → Chagee-app pickup · changes fulfilment

The order leaves delivery for in-app pickup: no rider and no platform commission. Its revenue moves channel; it is not added unless the customer buys more.

Chagee gains about **S$1.09** an order: S$5.90 × (30% commission − 3% payment) − S$0.50 reward. Assumptions, not Chagee's terms.

What exists today

CHECKED OCT 2026

GrabFood orders earn CHAGEE rewards once the membership is linked to CHAGEE on GrabFood; rewards are redeemable only on in-app pickup orders. In-app delivery exists where the address is in coverage. Whether Grab self-pickup orders earn is not stated.

Not assumed

Chagee cannot identify Grab customers or see their purchase history unless they link their account. Rewarding Grab self-pickup through membership is a **proposed integration**: account linking, transaction verification and reward rules must be confirmed.

VALUE | Delivery to app pickup gains about S$1.09 an order after the reward; Grab pickup to app pickup gains only if Grab's pickup commission is above 11%.

NEXT The business case and the manager's decision.

SPEAKER NOTES · REWARD CONVENIENT SELF-PICKUP

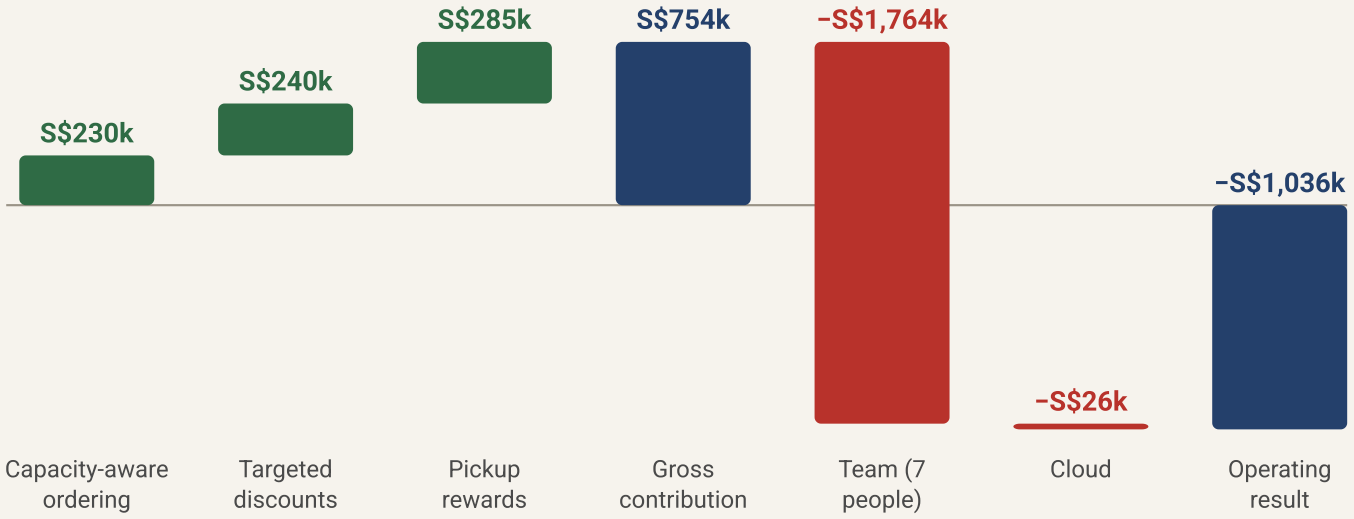
Initiative three, rewarding convenient self-pickup. Step through the checkout: the customer chooses pickup, sees transparent savings, and receives an eligible reward. Then switch journeys, because they are not the same conversion and do not save the same.

Delivery to Chagee-app pickup changes fulfilment and channel: Chagee avoids the platform commission, assumed 30% as on the expansion slides, less a 3% app payment cost and the reward, about S$1.09 an order on our assumptions. Grab self-pickup to Chagee-app pickup changes only the ordering channel: Chagee avoids Grab's pickup commission, which is not public; it pays only above about 11%. In both, the revenue moves channel; it is not added unless the customer buys more.

What we checked, in Chagee Singapore's Help Centre in October: GrabFood orders earn rewards once accounts are linked, rewards are redeemable only on in-app pickup orders, and the app now offers delivery where the address is in coverage. Whether Grab self-pickup orders earn is not stated. So the brief's idea of rewarding Grab pickup through membership is shown as a concept, a proposed integration, and is not in the totals. We never assume Chagee can identify Grab customers or see their histories. Context: in a Singapore choice experiment, consumers reacted more to menu-price premiums than to delivery fees (Ma and co-authors, 2024), and app adopters spent more in a US loyalty programme (Kim and co-authors, 2015). About one minute.

The initiatives add contribution, but not enough to fund the proposed team

A FULL-SCALE YEAR · SINGAPORE · OUR ASSUMPTIONS, NOT THE BRIEF'S



Base scenario. Moved orders shift revenue and count only the commission avoided, less rewards. Gross: S$128k low to S$1,158k high. Implementation spend S$880k; payback: none within 5 years.

Proposed plan

TEAM BRIEF · THREE PHASES, TWO GATES

Build and test	Pilot	Scale
months 1–6 · 5 people	months 7–12 · 7 people	months 13–18 · 7 people

Gate 1 · ready to pilot · month 6: forecasts and capacity checks reliable in shadow mode; app flows and control groups working; Grab linking confirmed or dropped

Gate 2 · ready to scale · month 12: incremental contribution after incentives and operating costs above the control group; pickup reliability and satisfaction hold

What would make it pay

- offers: at least **28%** of redeemed orders new
- Grab-pickup moves: pickup commission above **11%**
- run team: base contribution covers about **2** people, not 7

reconciliation: appendix →

MANAGER'S DECISION

Digital commerce manager
with CRM and operations

Which combination of pickup options, targeted offers and rewards increases completed orders and contribution while maintaining reliable service?

PROPOSED

Pilot all three in 3 outlets against matched outlets and a no-offer group; scale only past Gate 2, with a run team the contribution can carry.

MEASURE IN THE PILOT

- conversion and pickup reliability
- satisfaction; incremental purchases against a no-offer group
- contribution after incentives and operating costs

NEXT Once an order reaches the outlet, how should we organise the counter to serve it?

The business case, on the deck's own assumptions: 44 outlets, about 556 cups an outlet a day, S$5.90 a cup and 60% gross margin. Read the waterfall left to right. At base, rerouting keeps about S$230k of contribution a year, targeted offers add about S$240k after paying the discount on every redeemed order, and pickup moves save about S$285k after rewards: about S$754k gross contribution, S$128k to S$1,158k across our scenarios. Moved orders shift revenue; they are not new sales. Rerouted peak orders are not counted again in the counter chapter.

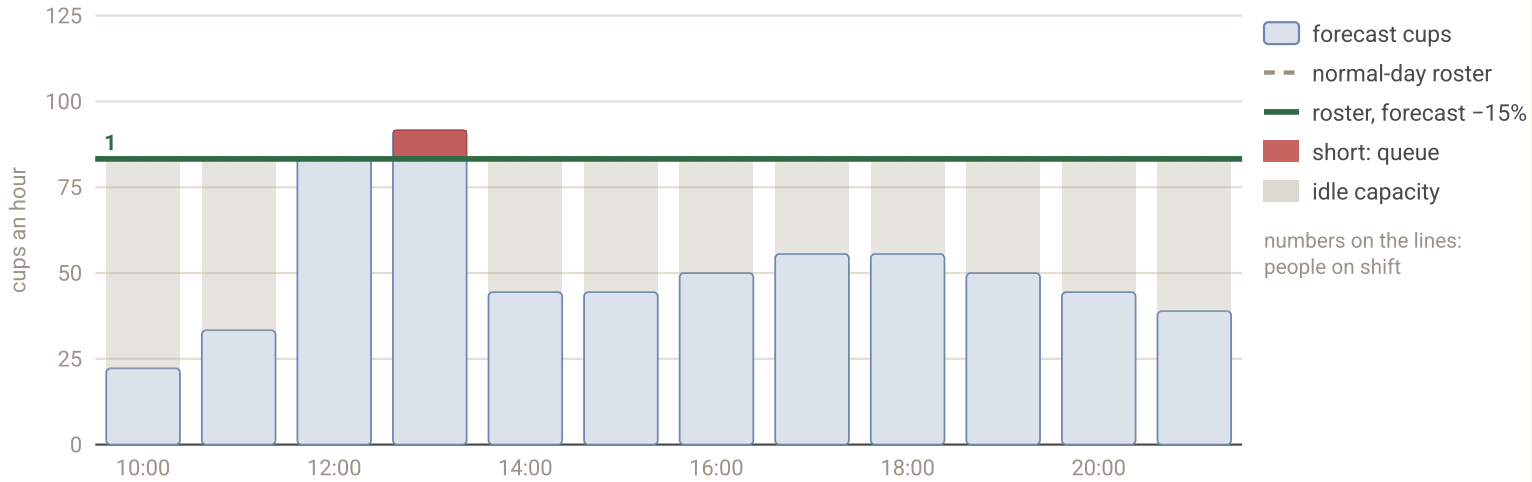
Then the costs. The brief's seven-person team, at the deck's approved specialist rate, costs about S$1,764k a year, plus about S$26k of cloud. So the operating result is about −S$1,036k a year at base, and the plan does not pay back within 5 years in any of our scenarios. Base contribution would carry a run team of about 2 people. That is why we do not repeat the brief's S$1.23m sales and S$0.89m profit: its arithmetic, reconciled in the appendix, counts discounts on new orders only, values pickup moves before rewards and leaves out the team.

The decision for the digital commerce manager, with CRM and operations: which combination of pickup options, targeted offers and rewards increases completed orders and contribution while keeping service reliable? Proposed: pilot all three in 3 outlets through the brief's three phases over eighteen months. Gate 1, at month six, checks readiness; Gate 2, at month twelve, requires incremental contribution after incentives and operating costs above matched outlets and a no-offer group before scaling. Then the transition: once an order reaches the outlet, how should we organise the counter to serve it? About one minute.

A forecast sets the crew each hour needs

BUSINESS PROBLEM Staffing and task allocation may not match changing demand, leaving queues, walk-aways or idle capacity. Observed people deployed on local expertise (Chagee source, anonymous); long reopening queues (Vulcan Post). Simulated everything below. To confirm Chagee's waits and walk-aways.

CUPS AN HOUR AGAINST CREW CAPACITY · BUSY DAY · ILLUSTRATIVE SIMULATION



SHOWN
Busy day, today's split, forecast −15% low. Other days: the deck's interactive version.

TAKEAWAY | Forecast −15% low: **1** at the peak, **20.8%** walk away, S$372 a day; peak-hour contribution **S$233** against S$287 with an exact forecast.

SPEAKER NOTES · DEMAND MOVES BY THE HOUR

Solution two, in-store management. The order has reached the outlet; now the counter has to serve it. The business problem: staffing and task allocation may not match changing demand. The solution: forecast staffing requirements and compare practical ways to organise people and machines. What we have observed: a Chagee source who asked not to be named told us outlet teams deploy people on local expertise, and Vulcan Post reported long queues at the Singapore reopening. What we have not observed: Chagee's own wait times and walk-aways. Everything on the chart is simulated, on our assumptions.

Step one is the forecast: cups an hour from historical order shares and the calendar. App signals, like the squad signals in the appendix, could refine it nearer the time, but they are not in these numbers. Step two: keep the normal-day roster on a busier day and the peak runs short, in red, while the morning crew stands idle, in grey. Step three: the forecast roster follows demand. Keep three things apart: the forecast says how many people; a sharing rule decides how they share the work; machines decide which steps people do at all. The next slide compares two ways for the same crew to share the work, then slide 10 compares automation setups; the bucket-brigade demonstration is in the appendix.

Step four, the sensitivity. On a busy day, a forecast 15% too low puts on 1 person at the peak instead of 2, and walk-aways go from 0.0% to about 21%. A forecast too high costs little: about S$15 more a day on an event day. The full table is behind Numbers. These error steps are scenarios, not Chagee's measured accuracy. About one minute.

Use the same crew more effectively during peak periods

Can the same crew serve peak demand better by changing how work is shared?

Order received
machine 2 s

→

Cup preparation
person 10 s

→

Dispensing
machine 10 s + person 12 s

→

Finishing and sealing
machine 8 s

→

Collection
person 8 s

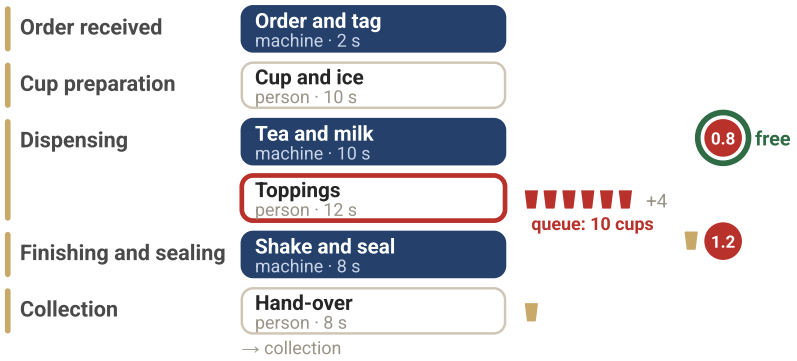
Illustrative counter setup, to validate with Chagee.
Hand-over is an assumed 8 s step, not a drink's make time. All by hand a cup is 60 s of work; here people do 30 s.

13:20:00 simulated · minute 20 of 60 · printed frame

Demand shown: 153 orders an hour.

A Fixed responsibilities

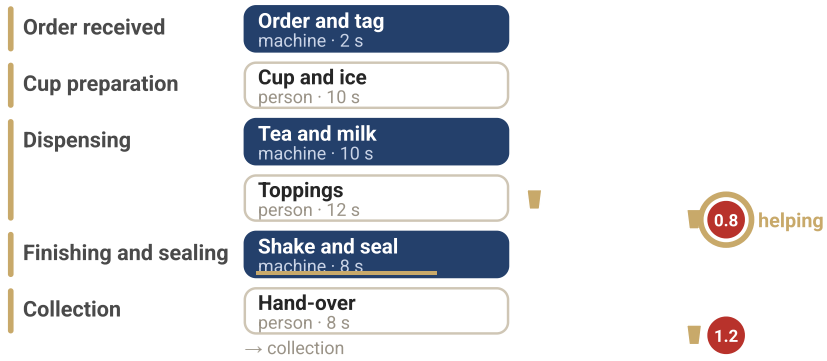
each person stays on their own steps · 2 people



orders placed **53** ready **35** in progress **12** walked away **6**

B Flexible assistance

a free person helps a queued step · 2 people



orders placed **53** ready **49** in progress **4** walked away **0**

ORDER-TO-READY · MEDIAN / 90TH PERCENTILE

A **4:10** / 6:28
B **2:04** / 4:18

ORDERS COMPLETED IN THE PEAK HOUR

A **134** of 151 · 11 unfinished · 7 away
B **144** of 151 · 7 unfinished · 1 away

PAID LABOUR FOR THE HOUR

A **S$30** · 2 people
B **S$30** · the same crew

Mean of 20 simulated peak hours at 153 orders an hour · orders placed 13:00–14:00, timed from order to collection · green: better.

TAKEAWAY | At 153 orders an hour the same 2 people, sharing flexibly: median 4:10 → **2:04**, 90th percentile 6:28 → **4:18**, 134 → **144** orders completed, same paid hour. A simulation result under our assumptions.

SPEAKER NOTES · SHARE THE WORK AT THE PEAK

The question for the operations manager: can the same crew serve peak demand better just by changing how the work is shared? The top strip is the counter as a sequence: order received, cup preparation, dispensing, finishing and sealing, collection, with machines on three steps and people on three. That split is our illustrative setup, to validate with Chagee; it is not Chagee's operation.

Be precise about the timing. The hand-over is an assumed 8-second step, not the time to make a drink. If every step were done by hand, a cup would take 60 seconds of human work in total; in this setup people do 30 seconds of it. Work content, the time between finished cups and the customer's order-to-ready time are three different measures: the last one includes waiting in queues.

Counter A keeps fixed responsibilities: each of the 2 people stays on their own steps, using the split that balances the work best. Counter B allows flexible assistance: when a person's own steps have nothing waiting, they walk to another hand step with at least 2 cups waiting and help, paying the walk and 3 seconds of coordination on every cup; they go back as soon as their own step has work. This is a separate rule from the bucket brigade in the appendix. Same crew, machines, orders and drink times on both sides.

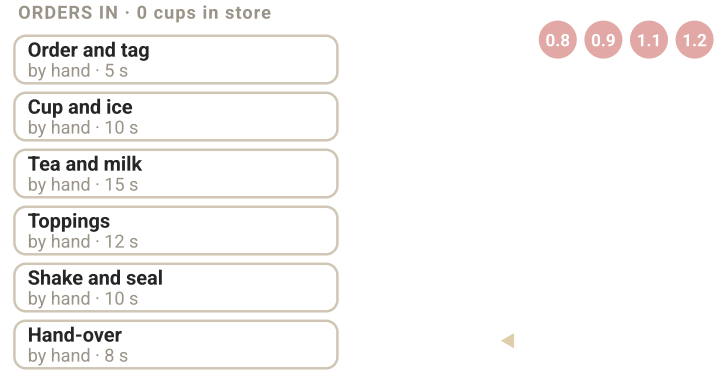
Play it at the event-day peak: on A, cups pile up at toppings while the colleague on cup and ice stands free. In the mean of 20 runs, B's median order-to-ready time is about 2:04 against 4:10, and its upper percentile 4:18 against 6:28, for the same paid hour. Move the demand to the normal-day peak: both rules perform the same, because no one is overloaded. These are simulation results under stated assumptions, not observations.

Forecast-informed staffing (slide 08) and machine investment (slide 10) are separate comparisons, so each improvement keeps its cause. The decision is on slide 11. About one and a half minutes.

In our model, one tea-and-milk machine is cheapest every day

PEAK HOUR · ILLUSTRATIVE SIMULATION

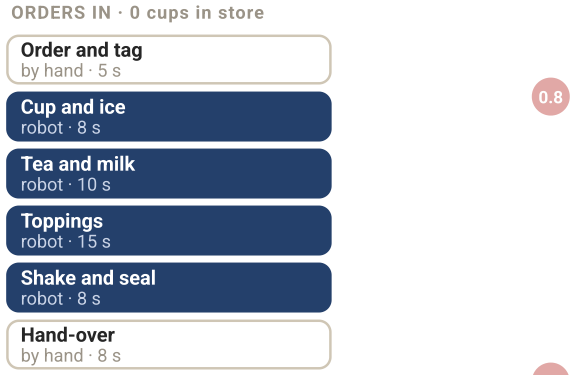
A All by hand 4 people, brigade



minute **20** / 60 served **57** walked away **0** hand-overs **156**

20 runs: away **0.0%** wait **1:25** **S$422**/day

B Machine-made drink (scenario) 2 people, fixed posts



minute **20** / 60 served **57** walked away **0**

20 runs: away **0.0%** wait **1:06** **S$465**/day

Navy = robot, white = by hand (click to switch) · red circle = a person · faded = walking · red cup = waiting for a person.

COST A DAY · EVENT DAY · WITHIN THE WALK-AWAY LIMIT

One machine (tea and milk) ✓	**S$335**
Today's split (our scenario)	**S$402**
All by hand	**S$422**
Machine-made drink (scenario)	**S$465**
All robots	**S$562**

✓ cheapest that keeps walk-aways within 2%. Full table: Numbers.

TAKEAWAY | One machine on tea and milk lets the roster drop a person off-peak for the cost of one unit. That holds only while a unit costs more than about S$905 a month; Chagee's real cost is unknown.

SPEAKER NOTES · SPLIT THE COUNTER

The forecast set how many people, and slide 09 how they share the work; this slide asks which counter steps go to machines. Side A opens on human-only: people do every step. Side B opens on machine-made drink: machines make the drink, people tag the order and hand it over. Both are our illustrative scenarios, not Chagee's disclosed setup. Chagee's own equipment cost is not public; its Singapore relaunch used semi-automated equipment, according to Vulcan Post.

Keep two things apart. Machine automation is the split, the equipment on each side. AI here is AI-assisted staffing: the demand forecast sets the crew. Numbers shows the AI layer on its own. Each side keeps its equipment, customers and sharing rule; only the crew moves from the normal-day roster to the forecast roster. On an event day, all by hand without AI has 2 people, and about 40% of peak-hour orders walk away; with AI the forecast puts on 4, walk-aways fall to 0.0%, median wait 5:32 to 1:24, for S$422 a day against S$361. On the machine-made drink the forecast keeps 1 person, so AI changes nothing there: 7.2% still walk away, and the split table adds a person to meet the limit. In our model, forecasting matters most where people do the work. The band follows the day and the equipment you pick; walk-aways and wait are the peak hour, mean of 20 runs; cost is the day's roster plus machines, without lost sales. Illustrative assumptions throughout.

How to use it: choose a split for each side, or click any station to hand it to a robot or a person, then run the peak hour or press End to compare. Each side starts with the fewest people, from the forecast roster up, that keep walk-aways within 2%.

The bars on the right compare five options on the chosen day, cheapest first; the full table is behind Numbers, mean of 20 runs: today's split, the machine-made drink, all by hand, all robots, and one tea-and-milk machine from the machine ladder in the appendix. In our model, one tea-and-milk machine is the cheapest of the five on every day type, because it lets the roster drop a

Roster to the forecast, test flexible sharing, pilot one machine

ONE LEVER PER COLUMN · PEAK HOUR · EVENT DAY · MEAN OF 20 RUNS

CRITERION	TODAY normal-day roster · today's split	+ FORECAST forecast roster · today's split	PROPOSED PILOT + ONE MACHINE forecast roster · tea-and-milk machine
Crew at the peak	1	2	**3**
Median wait	7:24	2:38	**1:44**
Walk-aways	53.8%	1.2%	**0.2%**
Cups served per person, peak hour	69	74	**50**
Peak-hour contribution	S$230	S$491	**S$482**
Cost a day: roster and machines	S$372	S$402	**S$335**
Flexible sharing, same crew: median ready time (slide 09)	one person, no one to help	4:10 → 2:04	**2:18 → 2:14**

READING THE TABLE

Green is the proposed pilot. Each column adds one lever: the forecast changes only the crew, the machine only the equipment. The upper rows share work by the bucket brigade (appendix); the last row compares the two slide 09 rules with that column's crew and machines. Illustrative simulation.

IF THE FORECAST IS OFF

A busy-day forecast 15% too low leaves about **21%** walking away (slide 08).

MANAGER'S DECISION

Operations manager / outlet GM

What staffing, task allocation and equipment best meet demand at an acceptable service level and cost?

PROPOSED, CONDITIONAL

In 3 outlets, try flexible assistance at the peak first; where queues remain, roster to the forecast or pilot one tea-and-milk machine. Simulated gains hold only under our assumptions.

MEASURE IN THE PILOT

- actual task times and equipment cycles
- who works which step, and movement limits
- order arrivals and order-to-ready times

NEXT After improving fulfilment across existing outlets, where would an additional store improve access and create value?

SPEAKER NOTES · THE OPERATIONS MANAGER'S DECISION

The decision for solution two belongs to the operations manager or outlet GM: what staffing, task allocation and equipment best meet demand at an acceptable service level and cost?

Read the table left to right; each column changes one lever. First, today: today's split and the normal-day roster. Second, the forecast changes only the crew. Third, a machine changes only the equipment, with the forecast crew. The upper rows share work by the bucket brigade, now in the appendix; the last row keeps each column's crew and machines and compares fixed responsibilities with flexible assistance, the slide 09 rules, so the sharing rule shows up on its own. That keeps task allocation, forecast staffing and machines apart.

The criteria are the ones the brief asks for: waiting time, walk-aways, productivity as cups served per person at the peak, and operating contribution. And the forecast will be wrong sometimes: on a busy day a forecast 15% too low lets about 21% walk away, while a forecast too high costs only about S$15 a day on an event day.

Our proposal is conditional. Try flexible assistance first, because it costs no extra paid hours: in the simulation it helps where one hand step queues while a colleague is free, as with the forecast crew on an event day, and changes little where no one is overloaded or where a machine already reshapes the work. Where queues remain, the fix is crew (roster to the forecast) or equipment (pilot one tea-and-milk machine against today's split), in 3 outlets. Every gain here is a simulation result under stated assumptions. The pilot must measure actual task times, equipment cycles, who works which step and how far they can move, order arrivals and order-to-ready times. Then the transition: after improving fulfilment across existing outlets, where would an additional store improve access and create value?

Open where an outlet adds customers, not where it moves them

BUSINESS PROBLEM Chagee needs to identify locations with sufficient additional demand and attractive economics: a new outlet may look successful while mostly moving sales from nearby stores. Context Chagee aims to expand rapidly in Singapore; the number of stores is confidential (Chagee source, anonymous). Simulated searches below.

1 WHO

Where should we open next?



Which location brings us new customers?
HEAD OF EXPANSION

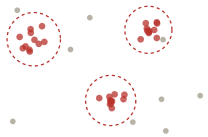
- **A** **Beside an existing outlet**: its sales may come from the store next door.
- **B** **Near underserved customers**: people searching with no outlet nearby.

The manager needs to compare locations, customer access and potential sales—not rely on intuition alone.

2 WHAT

Bring the signals together

- **App searches and unmet demand** — IN THIS DEMO · SYNTHETIC
- **Existing outlets and historical orders** — IN THIS DEMO
- **Residents, workers and transit access** — PROPOSED · NEEDS DATA ACCESS
- **Footfall and nearby competitors** — PROPOSED · NEEDS DATA ACCESS
- **Area delivery demand, if available** — PROPOSED · NEEDS DATA ACCESS
- **Rent and operating costs** — ASSUMED VALUES · P&L VIEW



DBSCAN: unsupervised machine learning that finds dense groups of searches. These clusters suggest locations to investigate. A separate demand model estimates sales and how much would shift from nearby outlets.

3 RESULT

Compare the options. Make the call.

This site looks promising. Let's compare the assumptions and downside before we commit.
HEAD OF EXPANSION



Site comparison synthetic app data · our assumptions

NET ADDITIONAL CUPS A DAY FOR CHAGEE

- **B** +731 cluster near Choa Chu Kang
- **A** +216 beside 34 · West Mall

SALES MOVED FROM NEARBY OUTLETS

B 106 of the 837 cups it sells · **A** 64% of its 604

DEMAND, DELIVERY AND COMPETITION hand-set scores, illustrative pin (appendix)

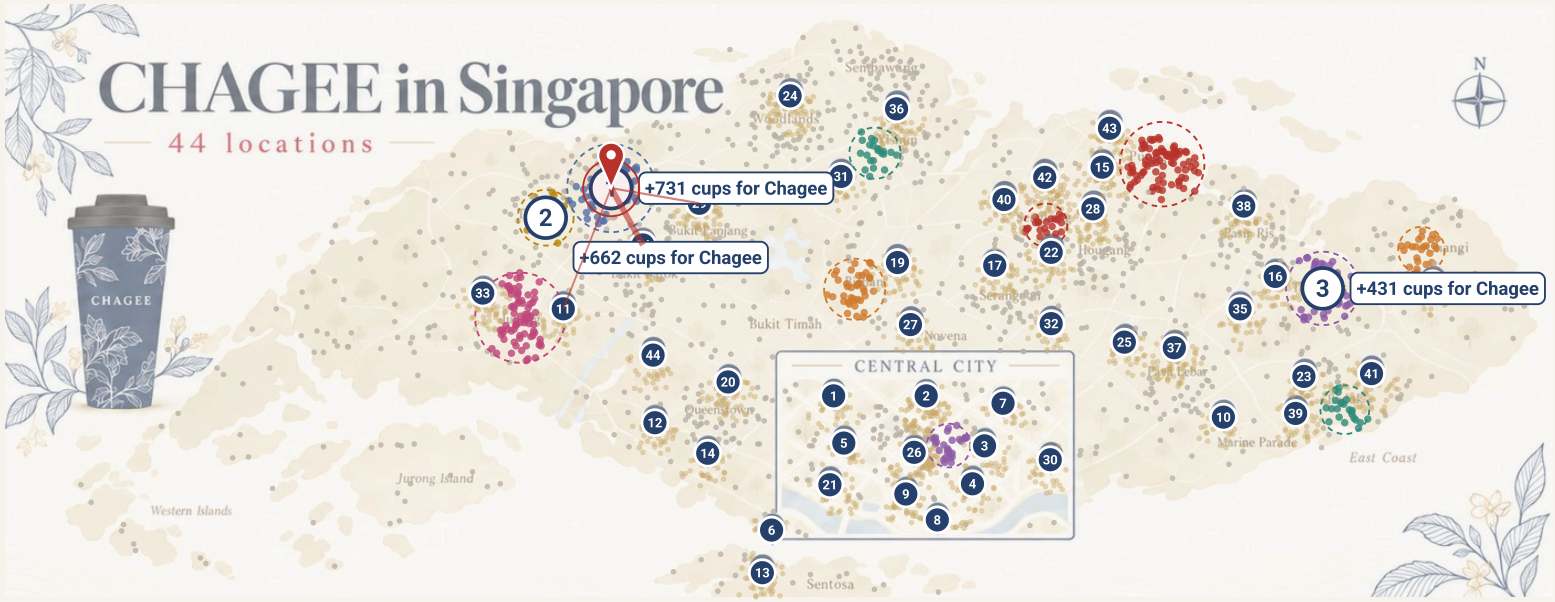
Demand | Delivery | Competition

An AI-assisted dashboard helps the team make an evidence-informed decision. The manager makes the final call.

Open where an outlet adds customers, not where it moves them

BUSINESS PROBLEM Chagee needs to identify locations with sufficient additional demand and attractive economics: a new outlet may look successful while mostly moving sales from nearby stores. Context Chagee aims to expand rapidly in Singapore; the number of stores is confidential (Chagee source, anonymous). Simulated searches below.

NEW-STORE FORECAST



SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL

Strong site: mostly new cups

+731 cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

34 · West Mall, 2.7 km	**−77**
29 · Bukit Panjang Plaza, 3.3 km	**−28**

Peak hour 13:00: **92 cups** → **2 people** on today's counter (slides 08 to 10).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method. Its five-year P&L: appendix.

TAKEAWAY | Judge a site by the cups it adds to Chagee, not the cups it sells: the cluster near Choa Chu Kang adds about **+731** a day; beside an existing outlet, the same store adds **+216**.

SPEAKER NOTES · PREDICT DEMAND FOR A NEW STORE

Solution three, new store expansion. The business problem: Chagee needs locations with enough additional demand and attractive economics. It has two sides: a new outlet can look successful while mostly moving sales from the stores around it, and customers elsewhere may have no convenient outlet at all. Context, from a Chagee source who asked not to be named: Chagee is looking to expand rapidly in Singapore, and how many stores it is going for is confidential. Chagee Singapore runs on its app, with few kiosks. If Chagee logs searches and their location, which we have to confirm, every search is a demand signal, including searches where no store is close. Here the searches are synthetic. That makes the app the forecasting input for a new store, before it has any sales.

Step one: synthetic app sessions on our map. Gold are within walking distance of an outlet; red found no store nearby. Step two: DBSCAN, unsupervised machine learning, groups red searches that are dense; scattered ones are noise. Step three: cluster centres are candidate sites.

Step four: click the map to test a site. The card leads with the gain for Chagee, which is the store's sales minus the cups it takes from nearby outlets, and gives a verdict: strong, mixed, or mostly moving cups. Try a pin next to Tampines West CC: it sells, but most of it moves from the two Tampines outlets nearby. Step five: the forecast's peak hour gives the opening roster for the counter on slides 09 and 10.

If asked how DBSCAN works, press How it works: it replays the algorithm one search at a time on a toy set. The appendix lists every simulated number.

The three sliders are the only inputs: cluster radius, minimum searches per cluster, and how far people will walk. The data is synthetic; with Chagee's real app logs, the same steps would run unchanged. About one and a half minutes.

Investigate the site that adds customers, then check the downside

THE SAME NEW OUTLET AT TWO SITES · SYNTHETIC APP DATA · FIVE YEARS, OUR ASSUMPTIONS

CRITERION	BESIDE 34 · WEST MALL OUR ASSUMED BASELINE	CLUSTER NEAR CHOA CHU KANG DBSCAN + DEMAND MODEL
Customer access: cups from people with no outlet within walking distance	47%	**98%**
Cups a day the store sells	604	837
Net additional cups a day, after cups moved from nearby outlets	+216	**+731**
Incremental operating profit, five years, base volume	−S$0.6m	**S$1.9m**
Investment at opening	S$0.4m	S$0.5m incl. AI share
Payback	not within 5 years	about 1.4 years
Downside: low volume, five years	−S$1.2m	S$0.03m, not paid back in 5 years
Upside: high volume, five years	−S$0.2m	S$3.6m

DBSCAN only finds where unserved searches gather; a separate demand model, on our assumptions, estimates sales. Synthetic searches. Operating profit before depreciation, interest and tax, after the contribution lost at nearby outlets. Every input and year: appendix, Expand.

MANAGER'S DECISION
Expansion manager

Which location should we investigate or pilot next?

PROPOSED

Investigate the cluster near Choa Chu Kang: rerun the method on Chagee's app logs before any lease or pilot.

PROCEED IF, ON REAL DATA

- most of its cups come from customers without convenient access
- net additional cups, not moved cups, carry the case
- profit covers the investment, with an acceptable downside

NEXT Our recommendation: three managers, three pilots.

SPEAKER NOTES · THE EXPANSION MANAGER'S DECISION

The decision for solution three belongs to the expansion manager: which location should we investigate or pilot next? The solution: identify underserved search clusters, evaluate candidate locations, and assess net additional demand and financial outcomes. The test: do its expected incremental returns justify the investment?

The table puts the same new outlet at two sites and reads down the criteria. Customer access: about 98% of the cluster site's cups come from people with no outlet within walking distance, against 47% beside the existing outlet. Net additional demand: about 731 cups a day against 216. Incremental operating profit over 5 years: about S$1.9m against −S$0.6m, on S$0.5m invested, paying back in about 1.4 years. Then the downside: at low volume the cluster site makes about S$0.03m and does not pay back within 5 years.

Say it plainly: DBSCAN does not predict sales or profit. It finds where unserved searches gather; a separate demand model, on our assumptions, estimates sales and cannibalisation, and the searches are synthetic. So the proposal is to investigate, not to open: rerun the method on Chagee's app logs before any lease. For a new market without app history, external area features and demand scenarios stand in: the optional extension after the close. About fifty seconds.

Three managers, three pilots, each judged on its own evidence

A connected story, not one combined model: no simulated benefit is carried between parts or added up.

ONLINE RETAILING · DIGITAL COMMERCE MANAGER

Which pickup options, offers and rewards?

RECOMMEND

Pilot capacity-aware pickup, targeted discounts and pickup rewards in 3 outlets, against matched outlets and a no-offer group, over the brief's three phases.

PROCEED IF

at Gate 2, completed orders and contribution after incentives and operating costs beat the controls, with reliable pickup and a run team the contribution can carry.

SERVE DEMAND · OPERATIONS MANAGER

What crew and counter setup should we use?

RECOMMEND

Roster to the forecast, test flexible sharing at the peak, and pilot one tea-and-milk machine against today's split in 3 outlets.

PROCEED IF

waits, walk-aways, cups per labour hour and contribution beat today's, even when the forecast is off.

EXPAND ACCESS · EXPANSION MANAGER

Which location justifies investment?

RECOMMEND

Investigate the cluster near Choa Chu Kang on Chagee's real app logs before any lease.

PROCEED IF

it adds net new customers, and its profit covers the investment with an acceptable downside.

TAKEAWAY | Coordinate online demand and pickup → serve demand reliably → open where we add new customers.

SPEAKER NOTES · OUR RECOMMENDATION

Our recommendation mirrors the three chapters: three decisions, three owners, and each one is a pilot judged on its own evidence.

For the digital commerce manager, with CRM and operations: pilot capacity-aware pickup, targeted discounts and pickup rewards in 3 outlets, against matched outlets and a no-offer group, through the brief's three phases and two gates. Proceed at Gate 2 only if completed orders and contribution after incentives and operating costs beat the controls, pickup stays reliable, and the run team is one the contribution can carry; in our model the seven-person team is not.

For the outlet GM or operations manager: demand moves by the hour, and the roster should too. Roster to the forecast, test flexible sharing against fixed responsibilities, and pilot one tea-and-milk machine against today's split. Proceed if waits, walk-aways, productivity and contribution beat today's, including when the forecast is wrong.

For the expansion manager: count the cups a new outlet adds, not the ones it sells. Investigate the cluster near Choa Chu Kang on Chagee's real app logs before any lease, and proceed only if its incremental returns justify the investment with an acceptable downside.

One caution to say out loud: these parts connect as a story, but we never add their simulated benefits together or count an order retained by online rerouting again in the counter or expansion numbers. About forty-five seconds.

THANK YOU

QUESTIONS AND DISCUSSION

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN



SPEAKER NOTES · THANK YOU

Recap the chain and its owners: the CRM manager makes membership worth joining, the operations manager serves the demand reliably, the expansion manager opens where demand remains underserved. Each part rests on its own assumptions; no simulated uplift is carried from one to the next.

Likely questions: is the brigade always better? No; with spare staff fixed stations are as good. Is the forecast real? No, it is an illustration of the chain, built on stated assumptions.

OPTIONAL EXTENSION

A NEW MARKET WITHOUT APP HISTORY: AUSTRALIA

WHAT WE CAN KNOW BEFORE OPENING · WHICH AREAS · WHICH SETUP · PILOT, LEARN, THEN EXPAND

SPEAKER NOTES · OPTIONAL EXTENSION: A NEW MARKET

Optional, after the close, for questions about a new market. It sits outside the three-part story and none of it feeds the main slides' numbers. Skip to the appendix with the agenda or the slide number box.

 New-market worked example · Australia · scenario assumptions

No Chagee sales history is not the same as no data: profile each area on what a new market can measure

Archetype candidate areas in one Australian state capital, standing in for real suburbs until the data is pulled. We assume Chagee has no operating history there.

CANDIDATE-AREA PROFILES · CONSISTENT FEATURES, EVERY AREA

FEATURE	STANDS IN FOR	A	B	C	D	A REAL VALUE WOULD COME FROM	VALUE HERE
Residents within 1 km	proxy walk-in demand from people living nearby	12k	18k	25k	20k	national census, usual residents by small area	assumption
Workers within 1 km	proxy weekday lunch and after-work demand	90k	20k	12k	5k	census place-of-work counts	assumption
Tertiary students within 1 km	proxy the young, app-first customer base	15k	40k	5k	2k	university and TAFE enrolment by campus	assumption
Household income index	proxy willingness to pay for a premium tea	115	80	95	90	census median household income	assumption
Transit access	proxy footfall past the counter	0.95	0.75	0.70	0.35	public-transport timetable data, services per hour at the nearest stops	assumption
Competitors within 500 m	proxy rival bubble-tea and cafe supply (lowers the score)	9	6	4	2	business listings or a map-places count	assumption
Rent, A$ a month	proxy local cost (lowers the score)	A$22,000	A$12,000	A$10,000	A$8,000	retail lease listings or an agent quote	assumption

Areas (archetypes): A CBD office core · B University precinct · C Suburban rail interchange · D Outer-suburb shopping centre

FOUR KINDS OF INPUT

Sourced observation none yet: no Australian data retrieved for this deck

Proxy what each feature stands in for, such as workers for weekday demand

Scenario assumption every value in the table

Unavailable before opening; left out, below

LEFT OUT · NEEDS CHAGEE ACTIVITY

- **Chagee app searches nearby**: needs Chagee app users in the market
- **Delivery orders from the area**: needs Chagee on the local delivery platforms
- **Distance to the nearest Chagee outlet**: there is none yet
- **Sales of existing Chagee outlets nearby**: there are none yet

Missing is unknown, not zero

No area has app searches or delivery orders yet. Set to zero, they would read as no demand; left out, the areas are compared only on what all of them have.

WHAT IS AI HERE Nothing yet. This is the data step the models on the next slides need; every value stays a scenario assumption until census and market data are pulled.

SPEAKER NOTES · NEW MARKETS: WHAT CAN WE KNOW BEFORE OPENING?

To consider: a new market. The main story was Singapore, where Chagee already trades. Now the harder case: a new market where Chagee has no local operating history. We use Australia as the worked example, and we assume no Chagee history there; that is to confirm.

No sales history does not mean no data. Before opening, a market still has residents, workers, students, income, transport access, competitors and rents. Step one: those are the features, and each is a proxy for something we care about, such as workers for weekday lunch demand. Step two: the same features for every candidate area, so the areas compare like for like.

Step three, be honest about the values. The areas are archetypes, a CBD office core, a university precinct, a suburban rail interchange and an outer-suburb shopping centre, not real suburbs. Every value is a scenario assumption. The table says where a real value would come from, such as the census, but we have not retrieved any Australian data, so there are no sourced observations on this slide yet.

Step four, what we leave out. App searches, delivery orders, distance to the nearest Chagee outlet and the sales of outlets nearby all need Chagee to be trading already. We leave them out rather than set them to zero: zero would read as no demand, when the truth is that we do not know. About forty-five seconds.

New-market worked example · Australia · scenario assumptions

B, A and C can each rank first as the weights move; the score shortlists areas, it does not forecast their sales

SUITABILITY SCORE OUT OF 100 · ILLUSTRATIVE WEIGHTS



B · University precinct rank 1–3 across 15 weightings · first in 6	**52**	cannibalisation 0%
A · CBD office core rank 1–3 across 15 weightings · first in 6	**52**	cannibalisation 0%
C · Suburban rail interchange rank 1–3 across 15 weightings · first in 3	**50**	cannibalisation 0%
D · Outer-suburb shopping centre rank 4 across 15 weightings · first in 0	**32**	cannibalisation 0%

Residents · Workers · Students · Income · Transit · Competitors · Rent · bar = score; rank range across the weight check

DEMAND SCENARIOS, CUPS A DAY · THE SAME FOR EVERY AREA

Low 278 Base 556 High 833

A placeholder scale: our Singapore per-outlet volume assumption, reused. With no calibration from real outlets, the score cannot turn into cups, so every area faces this range.

WEIGHTS USED

Residents 1.00 · Workers 1.00 · Students 1.00 · Income 0.50 · Transit 1.00 · Competitors 0.50 · Rent 0.50. Competitors and rent count against.

WEIGHT CHECK · 15 WEIGHTINGS

Each weight dropped, then doubled, one at a time. **B** first in 6, **A** first in 6, **C** first in 3, **D** first in 0. The top spot is not robust: treat the leaders as a shortlist.

WITH REAL OUTCOMES · SUPERVISED LEARNING

If outlets in existing markets had real sales, with these same features measured around each, a supervised model could learn the weights instead of us choosing them, tested on markets it has not seen. The supervised site model, also under Things to consider, is trained on synthetic Singapore data: **not validated for Australia**, and not used here.

Illustrative. Archetype areas and scenario assumptions: not Australian data, not Chagee data.

SPEAKER NOTES · AUSTRALIA CASE STUDY: WHICH AREAS SHOW POTENTIAL?

Same features, same definitions, every area. Step one: each feature is scaled across the 4 areas from worst to best, competitors and rent count against, and the weights add them up. At our weights the University precinct scores 52, the Outer-suburb shopping centre only 32.

Step two, and this is the point: move the weights. The check re-ranks 15 times, with each weight dropped and then doubled. B, A and C each come first in some of them; the Outer-suburb shopping centre is last in every one. So the score gives a shortlist, not a winner.

Step three: the score does not become a sales forecast. Turning it into cups needs a calibration from real outlets, which we do not have. So every area faces the same low, base and high range, 278, 556 and 833 cups a day, borrowed from our Singapore per-outlet assumption as a placeholder scale.

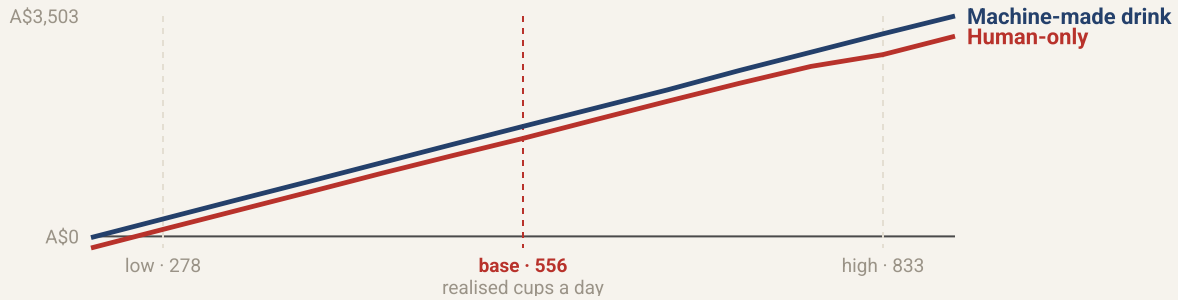
Where supervised learning fits: with real outcomes from outlets in existing markets, a model could learn these weights, tested on markets it has not seen. Our supervised site model is trained on synthetic Singapore data, so it is not validated for Australia, and we do not use it here.

Step four, cannibalisation. For the first opening it is zero: there is no Chagee outlet in any catchment. Switch to the next opening and it appears: an area within 5 km of the pilot loses up to 30% of its cups to it. About one minute.

New-market worked example · Australia · A$ scenario assumptions

With our A$ assumptions, machine-made drink holds up best in the worst case; human-only wins if a machine unit costs more than about A$3,258 a month

OPERATING CONTRIBUTION, A$ A DAY, YEAR AVERAGE · WITH AI STAFFING, EXACT FORECAST



EVENT DAY, BASE DEMAND (556 CUPS)	PEAK CREW	MEDIAN WAIT	WALK AWAY	A$ A DAY
Human-only · no machines				
Without AI · one crew all day	2	5:33	40.1%	A$1,766
With AI · forecast −30%	3	4:28	10.2%	A$2,139
With AI · forecast exact	**4**	**1:25**	**0.0%**	**A$2,218**
With AI · forecast +30%	5	1:08	0.0%	A$2,146
Machine-made drink · 4 machine units				
Without AI · one crew all day	1	5:51	5.4%	A$2,448
With AI · forecast −30%	1	5:51	5.4%	A$2,448
With AI · forecast exact	**1**	**5:51**	**5.4%**	**A$2,448**
With AI · forecast +30%	2	1:18	0.0%	A$2,491

Machines = automation: equipment sized before opening, held fixed. **AI = AI-assisted staffing**: the forecast sets the crew; realised demand unchanged. Event day, base demand · peak hour · 20 runs · illustrative

SHOWN

Event day, base demand, forecast exact in the band. Wage A$36, machine unit A$2,000 a month, rent A$12,000 a month (University precinct).

BREAK-EVEN, CUPS A DAY · AREA B

Human-only 268 with AI, 267 without; **Machine-made drink** 236 with AI, 235 without. Normal day, no walk-aways; low scenario 278.

PREFERRED SETUP · WORST CASE

Worst year-average day, any scenario and forecast error: human-only **A$108**, machine-made drink **A$276**. **Machine-made drink** is safer; the two cross at a machine unit of about **A$3,258** a month (now A$2,000); they do not cross within the wage range. With an exact forecast, machine-made drink leads at low, base and high demand.

WITHOUT AI INTELLIGENCE
one crew, all day

Human-only: 2 people · away **40.1%** · wait 5:33 · A$1,766/day
Machine-made drink: 1 person · away **5.4%** · wait 5:51 · A$2,448/day

WITH AI INTELLIGENCE
forecast roster

Human-only (forecast exact): 4 people · away **0.0%** · wait 1:25 · A$2,218/day
· +2 at the peak, +4 staff hours, +A$452
Machine-made drink (forecast exact): 1 person · away **5.4%** · wait 5:51 · A$2,448/day
· same crew: the forecast changes nothing here

SPEAKER NOTES · WHICH OPERATING SETUP WORKS UNDER UNCERTAIN DEMAND?

Now feed the demand scenarios into the same counter simulation as slides 09 and 10, with Australian costs in Australian dollars: a wage of A$36 an hour, A$2,000 a month per machine unit, A$8.5 a cup and A$12,000 a month rent at the University precinct. All scenario assumptions, none sourced. We compare human-only with machine-made drink. Equipment is sized before opening for the base scenario and does not change.

The chart is the year-average operating contribution, before set-up cost, delivery and overheads, as realised demand moves. Machine-made drink is above human-only across the range here, and both are below zero at the low end: break-even is about 236 cups a day for machine-made drink and 268 for human-only, against a low scenario of 278.

Then the AI layer, revealed in the band. Without AI the counter runs one crew all day, set once for a normal day's planned peak. With AI the forecast sets the roster for the day type, but the forecast is not perfect: it can miss realised demand by 30% either way. On an event day at base demand, human-only without AI loses about 40% of peak-hour orders; with an exact forecast, 4 people bring that to 0%, and the day's contribution rises to about A$2,218 from A$1,766. An under-forecast puts on 3 and gives some back: 10% walk away. On machine-made drink the forecast keeps 1 person even on an event day, so 5.4% still walk away: the staffing rule's own capacity estimate is the limit there, not the forecast.

When does the answer change? We judge by the worst case across demand scenarios and forecast errors. Machine-made drink keeps about A$276 a day, human-only A$108. That flips if a machine unit costs more than about A$3,258 a month; across our wage range, A$30 to A$45 an hour, machine-made drink stays ahead. Move the sliders to test it. Walk-aways are counted at the peak-hour rate over the busiest hours only, an approximation. About one and a half minutes.

EXTENSION · NEW MARKET · PILOT, LEARN, THEN EXPAND

New-market worked example · Australia · scenario assumptions

Pilot one machine-made drink counter in the University precinct, then let the pilot's own data decide the next step

RECOMMENDATION · CONDITIONAL

Where: the University precinct (B)

First at our weights; B, A and C are the shortlist. No own-store cannibalisation.

Setup: machine-made drink, AI staffing

Best worst case, A$276 a day against A$108 for human-only.

Holds only if

- a machine unit costs under about A$3,258 a month
- volume lands in the placeholder range, 278 to 833 cups a day; break-even is about 236
- the A$ wage, price and rent hold
- the forecast misses by no more than 30%

WHAT THE PILOT MEASURES, AND WHAT IT UPDATES

MEASURE	HOW	UPDATES
Orders, cups a day	till and app records, daily	the placeholder low, base and high range
Hourly demand	order times by hour, by day type	the hourly shape and peak the roster is built on
Conversion, where measurable	store-locator views or footfall against orders	the first link from score to cups
Service times	order to hand-over, per station	station seconds in the counter simulation
Labour needed	crew against waits and walk-aways each hour	the staffing rule and minimum crew
Machine cost and uptime	the lease quote, downtime log	the A$ machine cost and the setup choice

Then rerun: the area ranking with the area's real outcome, the setup comparison with real demand and times. Once several outlets trade, outcomes can train the supervised model; until then, weights stay a judgement.

EVIDENCE TO ACT · HELD 4 WEEKS IN A ROW

Example result, illustrative: 556 cups a day, 0.0% peak walk-aways, a quote of A$2,000.

- **Keep the pilot running.** 556 cups a day against about 236 (machine-made drink, University precinct).
- **No extra staff.** Peak walk-aways 0.0% against the 2% limit.
- **Keep machine-made drink.** Quote A$2,000 against the flip at about A$3,258 a month.
- **Open the next outlet, CBD office core.** It keeps 486 cups after 12.5% cannibalisation, against about 301 to break even.

EXPAND WHEN the pilot clears break-even for 4 weeks in a row, and the next area, the CBD office core, still clears it after losing about 13% of its cups to the pilot. Every figure here is a scenario assumption until the pilot replaces it.

SPEAKER NOTES · PILOT, LEARN, THEN EXPAND

So the answer to where, with what, and what next. Where: the University precinct, first at our weights, with B, A and C as the shortlist. With what: a machine-made drink counter with AI staffing, because it keeps the most in the worst case, about A$276 a day. It is conditional: it holds only while a machine unit costs under about A$3,258 a month, the A$ wage, price and rent hold, and volume lands in our placeholder range. Break-even is about 236 cups a day.

Step two, what the pilot measures: orders and cups a day, demand by hour and day type, conversion where we can measure it, service times per station, the labour each hour actually needs, and the real machine cost and uptime. Step three, how that updates the analysis: real volume replaces the placeholder range, real hourly demand replaces the Singapore shape, service times replace our station seconds, and the area's real outcome is the first calibration from score to cups. Once several outlets trade, those outcomes can train the supervised model.

Step four, the evidence to act, each held for 4 weeks in a row. Add staff when peak-hour walk-aways stay above 2% on the forecast roster. Revisit the setup if the firm machine quote is above about A$3,258 a unit a month. Open the next outlet, the CBD office core, only if the pilot clears break-even and the next site still clears it after losing about 13% of its cups to the pilot. Try a pilot result on the right. About one minute.

Appendix: the detail behind each chapter

Derivations, tables, sources and earlier analyses. Click a line to open it; use the agenda to return.

ONLINE RETAILING

Online retailing: business case and reconciliation

Online retailing: offers, plan, team and sources

GROW DEMAND

Members: when does joining feel worth it?

The proposed squad experience

Every group order helps us serve customers better

Pilot squads with personalised suggestions?

Members: the model and its checks

Members: break-even, assumptions and sources

Group orders to forecasts: method and results

ACROSS THE STORY

The data trail

Our scope: the earlier three-part framing

Chagee's data collection

Network effects across the three parts

Sources and assumptions

SERVE DEMAND

Bucket brigade: sharing the work

Counter comparisons, every day type

The modelled decisions in detail

The size of the benefit

The benefit in detail

Machine ladder

Machine ladder: tiers and sources

OPTIONAL EXTENSION

New market: inputs and method

EXPAND ACCESS

How the site forecast works

The new store in money: five-year P&L

Four scores for every location

The supervised model

From pin to net profit

SPEAKER NOTES · APPENDIX

The appendix holds everything moved off the main slides: derivations, the full tables, sources and earlier analyses. Click a line to jump to it.

Orders are recorded; the other forecast inputs are still to confirm

WHICH SOURCE FEEDS WHICH DECISION · STEP THROUGH

	Store and production	App and members	Delivery	Supply and context
Demand forecasts	Cups per hour by drink and channel	Member repeat demand	Orders per hour by platform	Weather, holidays, launches
Staffing and prep	Wait times, machine logs	Pickup no-shows	Hand-over peaks	Labour hours vs cups per hour
Replenishment	Drink mix per outlet	Launch trial rates	Group-order toppings	Ingredient use, expiry, par
Queue and capacity	Wait time, footfall vs sales	Store-select drop-offs	Rider wait at counter	Stock-outs at peak
Site selection	Sales density per outlet	"No store nearby", home and work	Distance and fee by postcode	Mall footfall, competitors
Menu and promos	Drink share, customisation	Trial and repeat, voucher uplift	Delivery basket mix	Collab calendar

White chips: core input; grey: refines it; red rows: our scope. Wait times, footfall, searches and location are to confirm.

TAKEAWAY | Orders by hour are recorded; searches, location, waits and walk-aways need confirming. Today, GM and outlet teams deploy people on local expertise (Chagee source, anonymous); we show how a forecast could support those calls.

SPEAKER NOTES · THE DATA TRAIL

Each step lights up one decision and the data that feeds it. Skip to the last step under time pressure: our scope is the three counter rows: forecast, staffing, queue.

Only the app ties a cup to a person, so any use of it needs consent and care under Singapore's data protection law (PDPA). This list is our reading of what Chagee's systems could record. Orders by channel and time are known; searches, customer location, wait times and walk-aways are candidates to confirm, based on public material and our industry conversation, not a Chagee data dictionary. About fifty seconds.

Grow demand, serve it well, then expand access

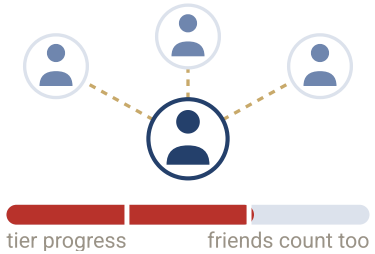
Three connected decisions, three owners. Each chapter: business problem → demonstration → manager's decision.

THREE CONNECTED DECISIONS

1

Grow demand

Make membership more rewarding.



CRM MANAGER

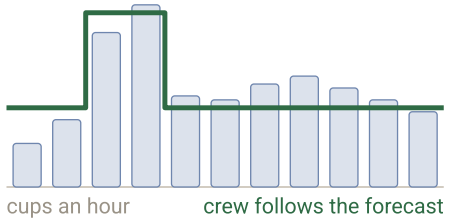
Which features and rewards should we pilot?

Appendix

2

Serve demand

Match staffing and workflow to demand.



OPERATIONS MANAGER

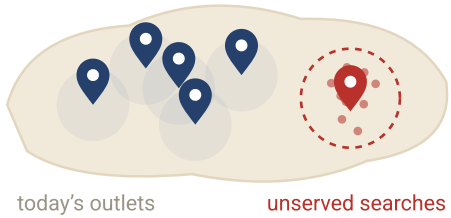
What crew and counter setup should we use?

Slides 08 to 11

3

Expand access

Open where we add new customers.



EXPANSION MANAGER

Which location justifies investment?

Slides 12 and 13

SPEAKER NOTES · OUR SCOPE

Our scope is three connected decisions, each with its own owner, and each chapter runs the same way: the business problem, a demonstration, then the manager's decision.

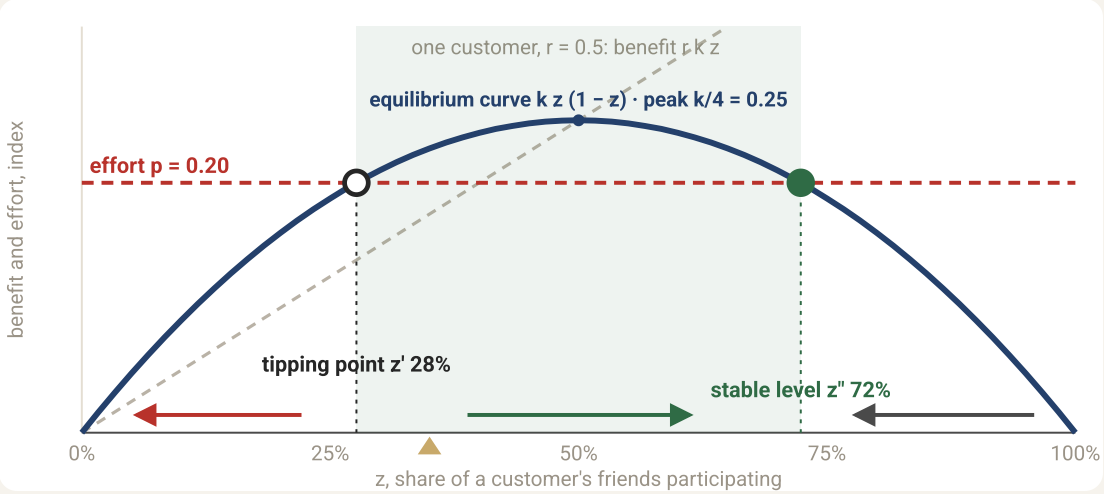
One, grow demand, for the CRM manager: make membership more rewarding. Which features and rewards should we pilot? Two, serve demand, for the operations manager: a more engaging membership has to be matched by reliable fulfilment, so match staffing and workflow to demand. What crew and counter setup should we use? Three, expand access, for the expansion manager: serving customers better also means being where demand is still unserved. Which location justifies the investment?

It is a connected story, not one model: we never carry a simulated membership uplift into the counter or expansion numbers. The tools each part uses come up in its own chapter. About forty seconds.

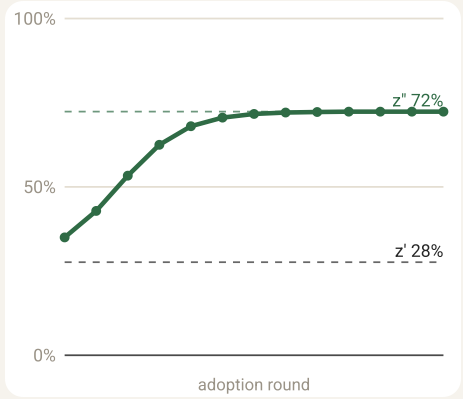
Squads add value only once enough friends take part

BUSINESS PROBLEM Chagee's Membership Operations team told us the problem they want to solve is **moving customers up the membership tiers**: customers may not think the effort of progressing is worthwhile.

BENEFIT AGAINST EFFORT · NETWORK-EFFECTS MODEL, NOT AI · ILLUSTRATIVE INDEX



PARTICIPATION, ROUND BY ROUND



SHOWN
Effort score 0.20 (an illustrative index, not money), value of friends k 1.00, starting participation 35%.

TAKEAWAY | Tipping point **28%**, stable level **72%**. Starting at 35%, participation rises to about **72%**. Illustrative.

SPEAKER NOTES · MEMBERS: WHEN DOES JOINING FEEL WORTH IT?

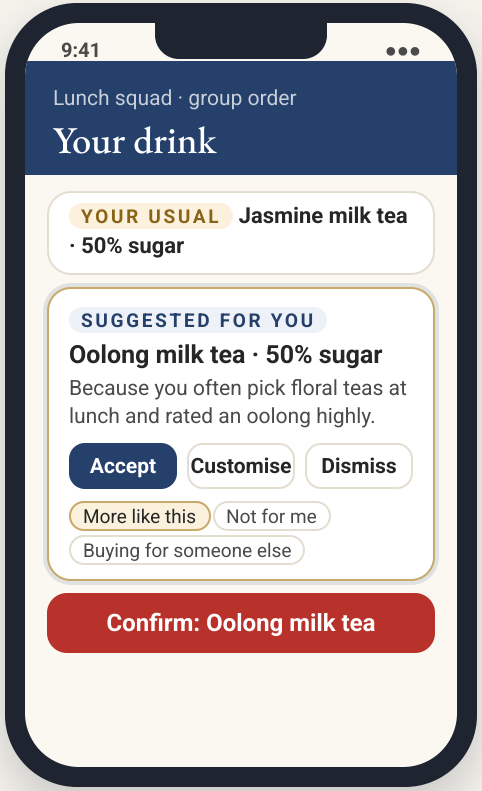
Part one starts with a business problem. Chagee's Membership Operations team told us the problem they want to solve is moving customers up the membership tiers: the barrier is whether customers think the effort of tiering up is worth it. We paraphrase them, and it is their view, not a measured finding: we have no data showing customers never move tiers. Our proposed answer is squad-based progression, where friends taking part make membership worth more. To see when that works, we use the network-effects model from Topic 8. It is an analytics model, not AI. One customer joins when the benefit beats the effort: interest times the value of friends times the share of friends on, against the effort score. Customers differ in interest, so the share who would join traces a hump.

Where the effort line crosses the hump there are two points. Below the lower one, the tipping point, participation falls back to zero; above it, it climbs to the stable level. At our default the tipping point is about 28%. Raise the effort and the two points close in, then vanish: the app stalls from any start. The symbols and equations sit behind How it works; the derivation and tables are in the appendix. About thirty seconds.

In the app, friends' orders count for everyone

PROPOSED EXPERIENCE · ILLUSTRATIVE PROTOTYPE

1. See your progress and start a squad
2. Invite friends from the share sheet
3. Choose your drink: your usual, or a suggestion
4. Each friend confirms their own drink
5. Collect the order: everyone's progress moves
6. Send a friend a drink
7. The friend picks from suggestions made for them



BASELINE AND AI

Recalling your last order is the baseline. The proposed AI part ranks other drinks from your own choices, acceptances and dismissals. Your feedback teaches it; "Buying for someone else" keeps that drink out of your profile.

WHY IT MATTERS

Easier choices may lower the effort, p, further. A hypothesis to test, like k.

Proposed experience with illustrative data and rules, not Chagee's; nothing is learned from real customers.

TAKEAWAY | At the members slide's effort of **0.20**, squads (assumed k 1.00 → 1.30) would move the tipping point from **28%** to **19%**: fewer friends need to be on before joining feels worth it. Assumptions to test, not measured effects.

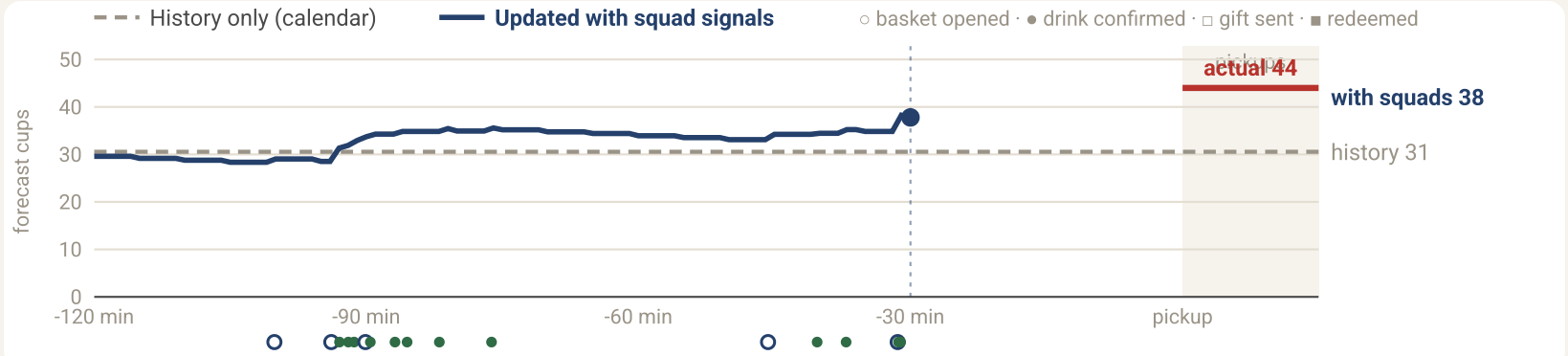
SPEAKER NOTES · THE PROPOSED SQUAD EXPERIENCE

This is the solution, as an illustrative prototype; the rules are ours, not Chagee's. Two parts, kept apart: squads, a product design, and personalisation, the AI part, which suggests a drink from each member's own choices and lets a gift's recipient choose privately. Click through: a member sees their progress and starts a squad, invites friends, then chooses a drink, with a suggestion they can accept, customise or dismiss and feedback such as not for me. In the shared basket each friend confirms their own drink, and preferences are saved to whoever chose, not whoever paid. When the order is collected everyone's progress moves. Last, gifting: the recipient picks privately from options ranked for them. Those picks and that feedback are the signals on the next slide.

Why it matters for the model: our hypothesis is that shared progress raises the value of friends, k, so the hump rises and the tipping point falls, from about 28% to 19% at the default effort. The stalled case is the headline: at a high effort the app never takes off without squads, and with them it can. These are assumptions to test, not measured effects. About thirty seconds.

Every group order helps the outlet prepare for demand

CUPS FOR THE 13:00 TO 13:30 PICKUPS · SYNTHETIC



Predict

38 cups expected at -30 min =

21 usual, from history + **13** confirmed in squads + **3.4** likely

Likely: invites not yet confirmed and gifts not yet redeemed, each counted at the chance it converted in past windows (21% and 19%), plus baskets not yet opened.

Act

Prepare
Jasmine 14 · Oolong 13 · Black 6
drinks to have ready

Staff
1 person, 13:00 to 13:30
the window's crew

Counter
one person, machines on tea and milk · load 76%
how the line is run

Pickup estimate
ready in about 3 min
shown to customers (rough)

SHOWN

Squad rush (synthetic), squad share 30%, forecast at -30 min, actual revealed. Window 13:00 to 13:30.

HOW TO READ IT

1. **Grey dashed**: what history alone predicts for these pickups.
2. **Navy**: the same forecast, updated as squad orders come in (dots under the chart).
3. **Act**: the outlet prepares, rosters and quotes pickup times from the navy number.
4. **Reveal actual**: see which forecast was closer.

TAKEAWAY | Squad rush: actual **44** cups. At -30 min the history-only forecast was off by **13.4**, the squad forecast by **6.2**: squad orders showed the rush coming before history could. Across 300 synthetic windows the squad forecast was closer on 67% of them.

The squad experience is also a data source. As the course put it for online retailers, an app can capture demand data, not just sales data. Press Play: squad signals arrive before one outlet's pickup window, and each forecast uses only what was known at that minute. What the app would record, and which of it is already documented, sits behind What the app records; the squad fields are proposed and need consent.

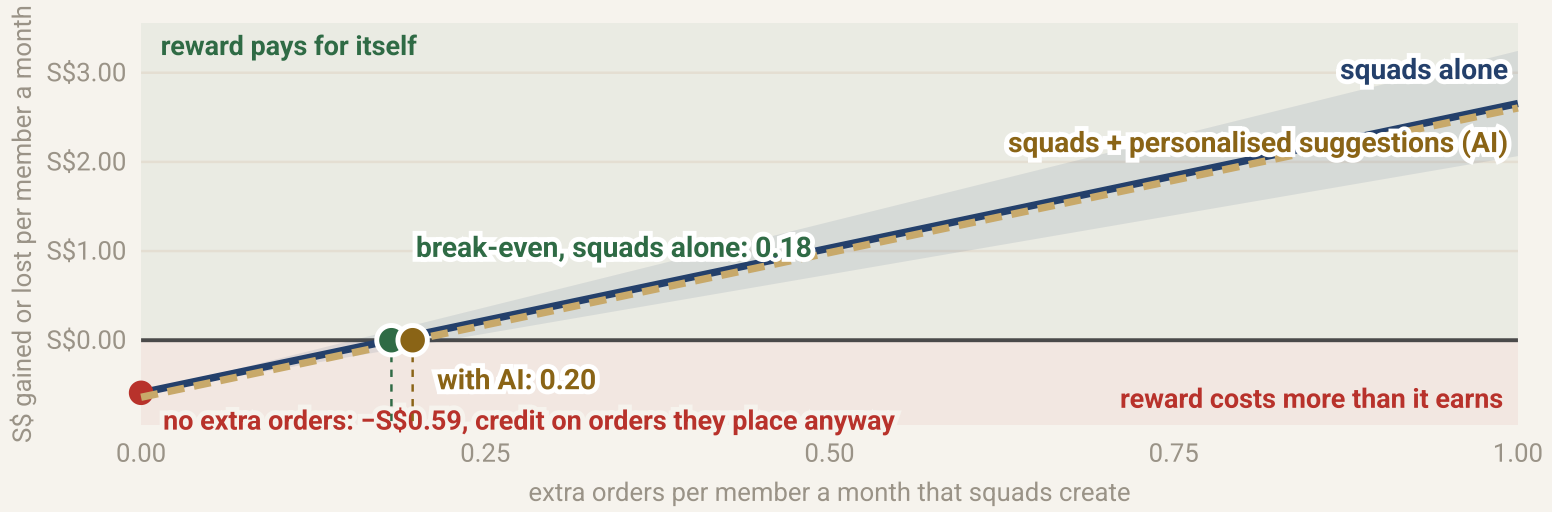
Step two, predict: expected cups are the cups already confirmed, plus a prediction of those not yet confirmed. An open basket or an unredeemed gift is not a drink; it counts only at the chance learned from past windows, and a confirmed cup is never counted twice. Step three, act: the forecast sets what to prepare, the crew for the window, how the counter is organised and the pickup time customers see.

Then play it. This is synthetic: the squad signals help only because we assumed they carry information, so watch the mechanics, not the score. On some days the enhanced forecast is worse, because walk-in surges are invisible to squads. Averaged over 400 synthetic windows it is off by about 5.3 cups ten minutes before, against 7.3 for the calendar baseline; with no squad share the two are identical. The appendix has the table.

Both sides gain: easier choices and reliable pickup for customers, a clearer view of what to prepare for Chagee. Personalisation may lower the effort on the members slide, but that is a hypothesis, and we do not turn adoption into extra orders. How the outlet organises people and machines to serve this demand is part two. First, the CRM manager's decision: is it worth piloting, and at what reward level? About one minute.

Pilot squads only at a reward their extra orders can pay for

DOES THE REWARD PAY FOR ITSELF? · AFTER REWARD AND AI COSTS · ILLUSTRATIVE



SHOWN

Credit 5.0% of the S$5.90 price; 2.0 group orders a month anyway; AI cost S$0.05 a member a month; margin 60% (band 50%–70%).

HOW TO READ IT

1. **Start**: no extra orders costs **S$0.59** a member a month, the credit on orders placed anyway.
2. **Rise**: each extra order earns its margin (band: 50%–70%).
3. **Break-even**: **0.18** extra orders a member a month, about one every 5 months.
4. **AI** adds S$0.05 a month, so **0.02** more.

MANAGER'S DECISION

CRM manager

Which features and rewards should we pilot?

PROPOSED

Three arms in Singapore (today's membership, squads, squads with suggestions) at a **5.0%** credit: squads must add **0.18** orders a member a month, suggestions **0.02** more.

PROCEED IF, ARM AGAINST ARM

- tier progression and orders rise against today's membership
- satisfaction holds
- contribution after reward and AI costs is positive

NEXT A more engaging membership must be matched by reliable fulfilment.

SPEAKER NOTES · PILOT SQUADS WITH PERSONALISED SUGGESTIONS?

The decision for part one belongs to the CRM manager: which features and rewards should we pilot? The line is the contribution per member a month after the reward, the group-order credit. It starts below zero, because the credit is also paid on orders members would have placed anyway, and rises with each extra order. At a credit of 5% squads alone break even at about 0.18 extra orders a month at our base margin. Push the credit up to the margin and it never breaks even.

Step four adds personalised suggestions. They cost something to run; we assume S$0.05 a member a month, a placeholder to replace with a real quote. That shifts the line down: break-even moves to about 0.20 extra orders, so personalisation must add about 0.02 orders a month of its own on top of squads.

That is why the pilot has three arms: today's membership, squads, and squads with personalised suggestions. Squads against today shows the squad effect; adding suggestions shows the personalisation effect. Proceed only if tier progression rises, satisfaction holds, the extra orders reach these break-evens, and contribution after reward and AI costs is positive. We measure orders directly and do not read purchases off the adoption model. Then the transition: a more engaging membership experience must be matched by reliable fulfilment. About forty-five seconds.

APPENDIX · ONLINE RETAILING · THE BUSINESS CASE, AND THE TEAM BRIEF RECONCILED (SLIDES 03 TO 07)

INPUTS: THE DECK'S OWN VALUES, AND WHERE EACH COMES FROM

Input	Value	Source
Outlets	44 (store directory, 15 Oct 2026). The store locator listed 46 on 24 Oct 2026: at that count every volume line is 5% higher	facts.sg_outlets; online_case.outlets_locator
Days, cups, price, margin	360 days; 200,000 cups an outlet a year (about 556 a day); S$5.90 a cup; 60% margin, so S$3.54 contribution a cup	store; storyboard
Channels	Chagee-app pickup and in-store app orders 60%; Grab delivery 30% at 30% commission; Grab self-pickup 5%; in-app delivery 5% (rider cost unknown, not valued)	site_ext (as on the expansion slides); online_case.channel_shares
Peak and off-peak	peak = the 4 busiest hours, 41% of cups; off-peak the rest	store.hourly_demand_share

EACH INITIATIVE, A FULL-SCALE YEAR: LOW · BASE · HIGH

Initiative	Eligible volume and rule	Contribution
Capacity-aware ordering	2,164,800 peak-hour Chagee-app cups (Grab orders cannot be rerouted); 1% · 3% · 5% kept × S$3.54	S$77k · S$230k · S$383k
Targeted discounts	3,115,200 off-peak app cups; 10% redeemed with S$1.00 off; 20% · 50% · 60% of them new. Contribution = new × S$3.54 − redeemed × S$1.00; break-even 28% new	−S$91k · S$240k · S$350k
Delivery → app pickup	2,640,000 Grab delivery cups; 5% · 10% · 15% move; each gains S$5.90 × (30% − 3%) − S$0.50 reward = S$1.09; S$1,558k of revenue moves at base, none added	S$144k · S$289k · S$433k
Grab pickup → app pickup	440,000 Grab self-pickup cups; each gains S$5.90 × (10% − 3%) − S$0.50 = S$-0.09; pays only if Grab's pickup commission is above 11.5%	−S$2k · −S$4k · −S$8k
Gross contribution	added sales (rerouting, plus new offer orders less every discount): S$184k · S$991k · S$1,430k	**S$128k · S$754k · S$1,158k**

THE TEAM BRIEF, LINE BY LINE: WHAT ITS FIGURES IMPLY, AND HOW WE TREAT THEM

Line	Brief	What it implies	Here, base
Base	46 × 500 cups × 365 days × S$6.00 = S$50,370,000	the locator's count; 365 trading days	S$51,920,000 on the deck's inputs
Rerouting	sales S$604,000; profit S$423,000	3% of all peak revenue (40%), Grab included; a 70% margin	app orders only, 60% margin: S$230k
Dynamic pricing	sales S$630,000; profit S$252,000	5% of S$12,600,000, which is 25% of the base: all delivery revenue, not off-peak, where app vouchers cannot reach; 40% margin; no discount on orders that would happen anyway	off-peak app orders, discount on every redeemed order: S$240k
Self-pickup	S$1,260,000 moves; profit S$214,000	10% of that delivery revenue × (20% − 3%); no reward cost; Grab pickup still pays Grab	two journeys, the deck's 30% commission, reward deducted: S$285k
Total	sales S$1,234,000; profit S$889,000	gross: no team, cloud or integration cost	gross S$754k; operating −S$1,036k after the run team and cloud
Payback	within a year if build costs stay below the annual profit	build cost against one year's gross; the team's running cost left out	phased cash flow: no payback within 5 years in any scenario
Scale	stores averaged over 1,500 cups a day, Q2 2025	could not be verified (transcript inaccessible); most likely group or China, not Singapore	not used

SPEAKER NOTES · ONLINE RETAILING: BUSINESS CASE AND RECONCILIATION

Backup for slide 07. Left: every input, with the deck's value and its source, and how each initiative's eligible volume and contribution are built at low, base and high. Right: the costs, the phased cash flow and the payback, then the team brief's figures line by line: what each implies and how we treat it. In short, its base uses 46 outlets at S$6.00 and 365 days; its pricing line applies a 5% uplift to all delivery revenue, which app vouchers cannot reach, and deducts no discount on orders that would have happened anyway; its pickup line values moves before any reward; and its payback leaves out the team. Soft copy only.

COSTS, CASH FLOW AND PAYBACK · BASE (S$; LOW AND HIGH IN BRACKETS)

Year	Benefit	Team	Cloud	Integration	Net	Cumulative
1	S$26k	−S$1,512k	−S$1k	−S$250k	−S$1,737k	−S$1,737k (−S$1,759k; −S$1,723k)
2	S$608k	−S$1,764k	−S$21k	–	−S$1,177k	−S$2,914k (−S$3,441k; −S$2,575k)
3	S$754k	−S$1,764k	−S$26k	–	−S$1,036k	−S$3,950k (−S$5,103k; −S$3,207k)
4	S$754k	−S$1,764k	−S$26k	–	−S$1,036k	−S$4,986k (−S$6,765k; −S$3,839k)
5	S$754k	−S$1,764k	−S$26k	–	−S$1,036k	−S$6,022k (−S$8,428k; −S$4,471k)

Implementation spend, months 1 to 6: S$880k (team and integration). Run cost from month 19: S$1,764k team (7 people × S$21,000 a person-month, the approved specialist rate) and S$26k cloud. Payback: none within the horizon. With a 3-person run team the base operating result would be −S$28k a year.

PROPOSED OFFER POLICY (TEAM BRIEF): DISCOUNTS ONLY, NOT PROVEN OPTIMAL

Scenario	Offer	Condition we add
Off-peak	up to S$1.00 on the highest-margin drink	only to members the model scores as likely to add an order
Peak	no vouchers; S$0.50 off a pre-order; limit Grab delivery when slots are full	the pre-order discount only if collection moves into a window with room; Grab limits as Grab's merchant tools allow (to confirm)
Weather, events	S$0.50 off in hot weather; S$1.00 in normal weather or rain; none during nearby events	weather and event effects estimated, not assumed
Inactive member	up to S$1.00 off their favourite for 7 days; a free upsize after 14 more days	only members the model scores as lapsing and responsive

PROPOSED PLAN: THREE PHASES, TWO GATES, AND THE TEAM AS IT JOINS

Phase	Months	Scope	People	Team cost
Build and test	1–6	capacity forecast by ten-minute window, offer engine, pickup reward flow; no-offer control group set up	5	S$630k
Pilot	7–12	pilot outlets against matched control outlets and a no-offer group	7	S$882k
Scale	13–18	roll out across Singapore outlets, ramping evenly	7	S$882k

Gate 1 · ready to pilot (month 6): forecasts and capacity checks reliable in shadow mode; app flows and control groups working; Grab linking confirmed or dropped; **Gate 2 · ready to scale** (month 12): incremental contribution after incentives and operating costs above the control group; pickup reliability and satisfaction hold. Roles: Product manager (1, from month 1), Data scientists (2, from month 1), App engineers (2, from month 1), CRM analyst (1, from month 7), Store-operations lead (1, from month 7).

SOURCES THE BRIEF CITES, CHECKED 24 OCT 2026

Source	Supports	Does not support
De Vries, Roy and De Koster 2018	longer waits relate to customers leaving and returning less (one Indian restaurant)	the 'nearly 15%' is a simulation without any waiting, not a Chagee uplift
Wang and co-authors 2024	pooled capacity for app and platform orders gives more profit and throughput, in a queueing model	a size for Chagee
Tong and co-authors 2020	platforms with dynamic pricing show higher demand (China O2O)	that targeted discounts lift Chagee's orders; an association
Ma and co-authors 2024	Singapore consumers more sensitive to menu-price premiums than to delivery fees (stated choices; partly verified)	observed Chagee behaviour
Kim, Wang and Malthouse 2015	app adopters spend more (US loyalty programme)	a Chagee uplift
Traynor and co-authors 2022; Liu and Li 2023	operators see commissions as a cost (US interviews); a model that assumes and caps commissions	Chagee's commission rate
CHAGEE SG Help Centre, Oct 2026	rewards redeemable only on in-app pickup; GrabFood orders earn once accounts are linked; in-app delivery within coverage; app payment methods	that Grab self-pickup orders earn; 'no delivery in the app' (the page says delivery is available)
Q2 2025 earnings call	could not be accessed	the '1,500 cups a day' figure; not used

Backup for slides 05 to 07: the brief's offer rules as a proposed policy, the three phases and two gates with the team as it joins and what it costs, and each cited source with what it does and does not support. Research findings come from other settings; they support the direction, not the size, of any Chagee effect. Soft copy only.

Chagee's data collection: what is public, what is to confirm

Only what our sources say; everything else in the deck is synthetic or illustrative.

- **Public sources**
  - Chagee's annual report: *"Our business generates and processes personal, transactional and other data."*
  - The app: *"Order and pay digitally"*, *"Order for self-pickup ahead of time"*, *"Locate our outlets and their operating hours"*.
  - After the Singapore reopening queues, *"the brand rolled out in-app ordering and delivery"*.

  · Chagee Form 20-F, FY 2025 · CHAGEE App Store listing · Vulcan Post, Oct 2024

- **Chagee source, anonymous**
  - Singapore outlets run under **one company and one data system**.
  - All delivery is through **third-party apps**, so delivery orders arrive by platform.
  - People are deployed on **local expertise**: the GM and outlet teams make the staffing calls.

  An internal staff member who asked not to be named; paraphrased, not quoted.

- **To confirm**
  - The **app's share** of Singapore orders: not in the parts of the annual report we read.
  - Whether **searches or location** are logged. The app's privacy label lists precise location as not linked to the user and does not list search history (storefront seen: not Singapore).
  - **Wait times, footfall and walk-aways**.
  - Any use of app data needs **consent** under Singapore's data protection law (PDPA).

TAKEAWAY | Orders are recorded in one system; searches, location and waits are to confirm. That is why our app and site data are synthetic (the group-orders slide and slide 12).

SPEAKER NOTES · CHAGEE'S DATA COLLECTION

Backup for slide 01's "Why Chagee?" and the data trail (appendix). Only three public sources speak to Chagee's data: the annual report's risk factors, the app's store listing and Vulcan Post on the reopening. Our Chagee source, who asked not to be named, added that Singapore runs on one company and one data system, that delivery is all third-party, and that staffing is decided by the GM and outlet teams. What we could not confirm: the app's share of orders, whether searches or location are logged, and waits or walk-aways; that is why the app and site data in the deck are synthetic.

From one customer's rule to the tipping point, and how far a friend network moves it

DERIVATION (OPIM639 TOPIC 8, SLIDES 30 TO 36)

Customer x in (0, 1) has interest r(x) = 1 - x. With a share z of friends participating, x joins if r(x) k z >= p, that is x <= 1 - p/(k z): the share who would join is 1 - p/(k z). An equilibrium is a share that reproduces itself, z = 1 - p/(k z), which gives **p = k z (1 - z)**. This hump is the equilibrium curve of the population; one customer's benefit is r k z, a straight line in z.

The hump peaks at k/4, at z = 50%. If p < k/4 the effort line crosses it twice: **z' = (1 - sqrt(1 - 4p/k))/2**, the unstable tipping point, and **z" = (1 + sqrt(1 - 4p/k))/2**, the stable level. At p = k/4 they merge at 50%; above it there is no positive equilibrium and participation falls to zero from any start. Rounds: z_next = max(0, min(1, 1 - p/(k z))).

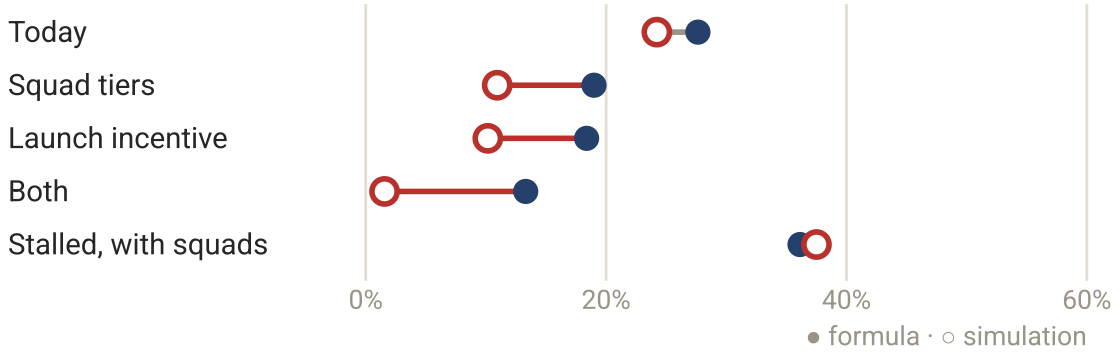
PARAMETERS (ILLUSTRATIVE INDEX, NOT CHAGEE DATA)

r individual interest, 0 to 1, uniform across customers. **k** value of friends participating; squads are hypothesised to raise it. **z** share of a customer's friends participating. **p** effort score of joining or progressing, 0 to 1: 1 is the whole benefit a fully interested customer gets with every friend on at k = 1; not money. Slider defaults: p 0.2, k 1, incentive 0, z0 0.35. Levers, read from the scenario table: squads add 0.3 to k; a launch incentive takes 0.05 off p.

SCENARIO TABLE, COMPUTED (SIM/MEMBERS.JS); THE BUILD FAILS IF IT DIFFERS FROM MEMBERS.EXPECTED_TABLE

Scenario	p	k	4p/k	z'	z"
Today	0.20	1.0	0.80	28%	72%
Squad tiers	0.20	1.3	0.62	19%	81%
Launch incentive	0.15	1.0	0.60	18%	82%
Both	0.15	1.3	0.46	13%	87%
Stalled app	0.30	1.0	1.20	none	none
Stalled, with squads	0.30	1.3	0.92	36%	64%

NETWORK CHECK · CLOSED FORM AGAINST THE FRIEND-NETWORK SIMULATION



Scenario	Closed form z'	Simulated	Gap, points
Today	28%	24%	-3.4
Squad tiers	19%	11%	-8.1
Launch incentive	18%	10%	-8.2
Both	13%	2%	-11.7
Stalled app	none	none	
Stalled, with squads	36%	38%	+1.4

2,000 customers on a seeded random graph with about 8 friends each, interest uniform; a customer is on next round if r k (share of friends on) >= p. 20 seeded runs; the simulated tipping point is the smallest starting share (to 0.01) from which the mean run ends above 10%. Gaps above five points mean the friend-group structure matters: with a handful of friends each, pockets of friends keep each other on, so the network tips lower than the formula, which assumes everyone sees the whole population's share. We report the gap, not hide it.

SPEAKER NOTES · MEMBERS: THE MODEL AND ITS CHECKS

Soft copy only. Backup for the members slides: the derivation, every parameter, the computed scenario table and the network check.

The break-even across the margin range, every new input, and what the sources say

BREAK-EVEN OF THE GROUP-ORDER CREDIT

Contribution per member a month = price x (margin x E - credit x (B + E)), with E extra orders and B group orders a member would place anyway. Break-even E* = credit x B / (margin - credit); none if the credit reaches the margin. Price S$5.90, credit 5%, B 2, E 1.

Gross margin	Extra orders needed a month	With AI cost (CRM manager's decision)	Contribution a member a month at E
50%	0.22	0.24	S$2.06
60%	0.18	0.20	S$2.65
70%	0.15	0.17	S$3.24

With personalised suggestions (the CRM manager's decision) a flat AI cost a of S$0.05 a member a month is added: E* = (credit x B + a / price) / (margin - credit). Personalisation must add a / (price x (margin - credit)) = 0.02 orders a month on top of squads alone at the base margin.

NEW INPUTS (ASSUMPTIONS.JSON MEMBERS; ASSUMPTION, NOT CHAGEE DATA)

group credit share: 0.05 (range 0, 0.1). New input, assumption, not Chagee data.

extra orders per member month: 1 (range 0, 3). New input, assumption, not Chagee data.

base group orders per member month: 2 (range 0, 6). Invented for the break-even (the credit is paid on orders that would have happened anyway); assumption, not Chagee data.

ai cost per member month: 0.05 (range 0, 0.3). Illustrative placeholder, not sourced: the running cost of personalised suggestions per participating member, allocated. Replace with a quoted cost before any decision. Used only on slide 08 to show what personalisation must add on top of squads.

squad size: 2, 6. Our proposal, a design choice, not a Chagee feature claim.

example interest: 0.5. Display only: a middling customer, to separate one customer's benefit from the equilibrium curve.

app rules: points own drink 10, points per squad drink 5, points now 220, points next 300, tier now Silver, tier next Gold, people Mei, Jun, Aisha, Ravi, squad name Lunch squad, prefs Mei usual Jasmine milk tea · 50% sugar, suggest Oolong milk tea · 50% sugar, why you often pick floral teas at lunch and rated an oolong highly, Jun usual Black milk tea · less ice, suggest Black milk tea · 25% sugar, why you switched to less sugar in your last few orders, gift options Black milk tea · 25% sugar, Oolong milk tea · less ice, Fresh fruit tea, Aisha usual Fresh fruit tea, suggest Jasmine milk tea · no ice, why you liked jasmine teas when ordering for yourself, Ravi usual Oolong milk tea, suggest Oolong milk tea · less sugar, why close to your usual, with the sugar level you picked last week. Our illustrative rules for the proposed experience, used consistently on slide 06. Not Chagee's membership rules, which are to confirm. The group-order credit is members.group_credit_share. Preferences and suggestions are illustrative data, not learned from anyone. Recalling the last order is the baseline; the proposed AI part ranks alternatives from each member's own choices, acceptances and dismissals. No real messages, purchases or accounts.

network: customers 2000, friends per customer 8, runs 20, max rounds 200, z0 step 0.01, takeoff share 0.1. Assumption: eight friends each, a squad plus a few more. The closed form assumes everyone sees the whole population's share; a friend group sees a noisy sample of it.

SOURCE EXCERPTS · QUOTED WORD FOR WORD

> "we cannot accurately predict the rate or range of consumers who will join our membership program"
>
> Chagee Form 20-F, FY 2025, Item 3.D, risk factors

> "we may not be able to use our membership program to deepen our understanding of our target consumers"
>
> Chagee Form 20-F, FY 2025, Item 3.D, same risk factor

> "The value z′ is the hump the firm must get over in order to succeed."
>
> OPIM639, Topic 8, Service Platforms, slide 34

MEMBERSHIP OPERATIONS TEAM, CHAGEE

Paraphrased, not quoted: the barrier to levelling up is not outlet operations; it is whether customers think the effort of tiering up is worth it.

SPEAKER NOTES · MEMBERS: BREAK-EVEN, ASSUMPTIONS AND SOURCES

Soft copy only. Backup for the CRM manager's decision, and the source excerpts behind the chapter.

How the enhanced forecast is built, and how it compares on synthetic windows

WHAT THIS IS

A synthetic demonstration of the mechanics only: one outlet's peak half-hour pickup window, with squad signals generated by our assumptions. The enhanced forecast can only help to the extent we assume the signals carry information, so this shows how the forecast would work, not measured accuracy. Nothing here feeds any other slide.

THE FORECASTS

Baseline: the calendar expectation for the window, cups a day x the peak hour's share x half an hour. **Enhanced**: the baseline's non-squad part, plus cups confirmed so far, plus predicted cups not yet confirmed: open baskets' unconfirmed invitees x the chance learned from synthetic history that they still confirm by then, unredeemed gifts x the learned chance of redemption, and the squad cups history says arrive from baskets and gifts not yet started. Each forecast uses only signals up to its time; a confirmed cup is counted once; an invitation or an unredeemed gift is never a confirmed drink.

MEAN ABSOLUTE ERROR, CUPS, OVER 400 SYNTHETIC WINDOWS

Squad share	120 min before: baseline / enhanced	60 min before: baseline / enhanced	30 min before: baseline / enhanced	10 min before: baseline / enhanced
0%	6.75 / 6.75	6.75 / 6.75	6.75 / 6.75	6.75 / 6.75
15%	6.86 / 6.83	6.86 / 6.28	6.86 / 6.16	6.86 / 5.95
30%	7.29 / 7.23	7.29 / 6.11	7.29 / 5.65	7.29 / 5.31
45%	7.79 / 7.70	7.79 / 6.13	7.79 / 5.19	7.79 / 4.63

With no squad share the two are identical. The enhanced forecast improves only because the synthetic squad signals were generated to move with demand; on single days it can be worse, when walk-in demand moves and squads cannot see it. Not measured accuracy.

SIGNALS AND WHERE THEY STAND

Documented: completed purchases (Chagee's Form 20-F: the business generates personal, transactional and other data; L-sec20f-data); in-app ordering at a chosen outlet (Vulcan Post, P-vulcanpost-comeback; App Store listing, P-appstore, not checked by us). **To confirm**: drink customisations and the pickup time chosen, in Chagee's Singapore app. **Proposed, needing collection and consent**: group size and confirmed quantities, suggestion accepted, edited or dismissed, gifts sent and redeemed, open baskets and invitations.

INPUTS (ASSUMPTIONS.JSON SQUADCAST; ASSUMPTION, SYNTHETIC, NOT CHAGEE DATA)

window minutes: 30 minutes in the pickup window. The window is the peak hour's first half (store.hourly_demand_share), so its calendar baseline is the deck's own volume: cups a day x the peak hour's share / 2.

lead minutes: 120 minutes before the window that the playback starts. Display and simulation horizon.

social share: 0.3 share of the window's cups that come through squad group orders and gifts, on average (range 0, 0.5). Hypothesis to test; zero means no squad signals, and the enhanced forecast then equals the baseline.

invites per basket: 4 people invited to a group basket, on average (squad size 2 to 6) (range 2, 6). Illustrative.

member conversion: 0.75 share of invited members who confirm a drink (range 0.3, 1). Illustrative; the forecast does not know it and estimates it from synthetic history.

confirm delay minutes: 8 average minutes from a basket opening to a member confirming. Illustrative.

gifts per window: 1.5 gifts sent for this window, on average, at the default squad share (scales with the squad share). Illustrative.

gift redeem share: 0.6 share of gifts redeemed for this window. Illustrative; an unredeemed gift is never counted as a drink.

gift redeem delay minutes: 30 average minutes from sending a gift to its redemption. Illustrative.

day shock sd: 0.2 spread of the day's demand around the calendar baseline (both walk-in and squad cups move with it). Illustrative: why the calendar baseline misses on any one day.

history windows: 400 past synthetic windows the enhanced forecast learns its conversion and late-arrival rates from. Synthetic history, same generator, different seeds.

eval windows: 400 synthetic windows used to compare the two forecasts (appendix). Different seeds from the history.

eval leads: 120, 60, 30, 10 minutes before the window at which the forecasts are compared. Appendix table.

day picks: 0.8, 1, 1.25 actual cups against the calendar baseline for the three example days on the slide (quiet, typical, busy). Display: the example days are the seeds whose actual cups, as a share of the baseline, are closest to these.

success pick: 1.45 actual cups against the calendar baseline for the fourth example day on slide 07 (squad rush). Display: a success case chosen by rule, not by hand. Among the same 300 synthetic windows, only those where the squad forecast is closer than the history-only forecast at the slide's default minute; of those, the one whose actual cups are closest to this share of the baseline. The slide also states the share of all windows where the squad forecast was closer, so the case is not presented as typical of every day.

drink mix: Jasmine milk tea 0.35, Oolong milk tea 0.3, Black milk tea 0.2, Fresh fruit tea 0.15 share of cups by drink, for predicted cups not yet chosen. Illustrative generic drinks, not Chagee's menu or sales mix.

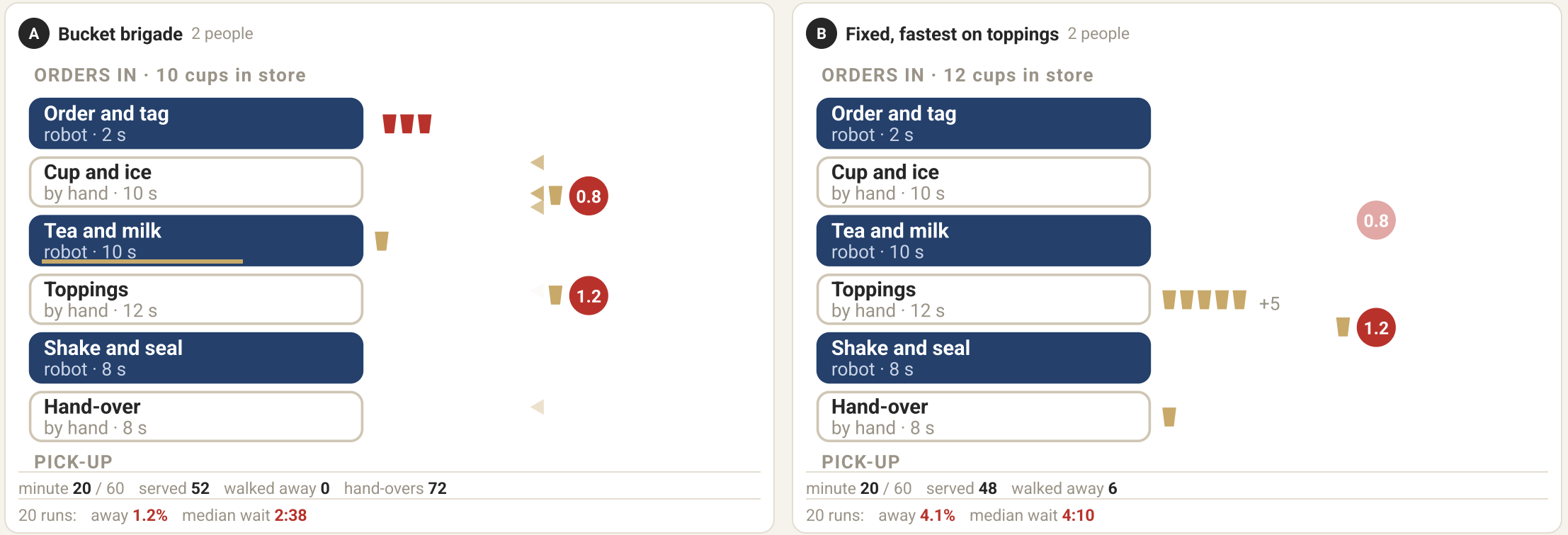
Course link: OPIM639 Topic 1, slide 12 (capture demand data, not just sales data) and slide 13 (personalised service); Topic 2, slide 17 (the operator that provides fulfilment holds the detailed data). C-topic1, C-topic2.

SPEAKER NOTES · GROUP ORDERS TO FORECASTS: METHOD AND RESULTS

Soft copy only. Backup for the group-orders slide: the forecast definitions, the synthetic comparison and every input.

A bucket brigade balances a lean crew by itself

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES · ILLUSTRATIVE SIMULATION



Red circle = a person (number = speed) · gold ring = held back · faded = walking · ▲ = hand-over · navy = robot · red cup = waiting for a person

TAKEAWAY | With a lean crew, work flows to whoever is free, so the line balances itself. With spare people, well-assigned fixed posts are as good or a little faster.

SPEAKER NOTES · SHARE THE COUNTER

The forecast on slide 08 set how many people. This slide is how they share the work: same counter, same customers, two sharing rules side by side. The counter is today's split, our scenario, not Chagee's disclosed setup. Robots, the navy boxes, take the order, brew tea and milk, and shake and seal. People do cup and ice, toppings and hand-over.

Left, a bucket brigade: each person carries a cup forward; the last hands it over, walks back and takes over a colleague's cup. Right, fixed stations: each person owns a station.

Press play on an event day. Then set the right side to "slowest on toppings": the queue builds at toppings and customers walk away. The brigade never needed to know who was fast.

Be honest about the other case: add people and fixed stations catch up, or are a little faster. About one minute.

If asked what the forecast changes here, open Numbers: the band there shows the AI layer on its own. Machines are automation, and here they stay exactly as they are: today's split on both sides. AI means AI-assisted staffing: the demand forecast sets the crew. Equipment, customers and each side's sharing rule are held fixed; only the crew changes. Without AI, the normal-day roster of 1 faces an event day: about 54% of peak-hour orders walk away with a brigade, 54% with fixed posts. With AI, the forecast crew of 2 brings that to 1.2% and 4.1%, for S$402 a day against S$372. The band follows the day you pick. Walk-aways and wait are the peak hour, mean of 20 runs; cost is the day's roster plus machines and leaves out lost sales. All illustrative assumptions.

Course link: the bucket-brigade simulation from Week 5 and Week 6, equipment and technology, inspired this counter replica. Same idea, a different use case.

Counter comparisons for every day type

SLIDE 08 · FORECAST ERROR: THE ROSTER FOLLOWS A WRONG FORECAST, THE CUSTOMERS DO NOT (TODAY'S SPLIT, BUCKET BRIGADE)

Forecast	Normal day crew · away · S$/day	Busy day crew · away · S$/day	Event day crew · away · S$/day
30% low	1 · 0.9% · S$372	1 · 20.8% · S$372	2 · 1.2% · S$402
15% low	1 · 0.9% · S$372	1 · 20.8% · S$372	2 · 1.2% · S$402
exact	1 · 0.9% · S$372	2 · 0.0% · S$402	2 · 1.2% · S$402
15% high	1 · 0.9% · S$372	2 · 0.0% · S$402	3 · 0.0% · S$417
30% high	1 · 0.9% · S$372	2 · 0.0% · S$402	3 · 0.0% · S$432

BUCKET BRIGADE (APPENDIX) · SHARING RULES AT THE FORECAST CREW, TODAY'S SPLIT

Rule	Normal day crew · away · wait	Busy day crew · away · wait	Event day crew · away · wait
Bucket brigade	1 · 0.9% · 2:08	2 · 0.0% · 1:14	2 · 1.2% · 2:38
Fixed posts, fastest on toppings	1 · 1.0% · 2:10	2 · 0.0% · 1:14	2 · 4.1% · 4:10
Fixed posts, slowest on toppings	1 · 1.0% · 2:10	2 · 0.7% · 3:26	2 · 34.9% · 9:47

SLIDE 09 · SAME CREW OF 2, FIXED RESPONSIBILITIES AGAINST FLEXIBLE ASSISTANCE, AT EACH DAY TYPE'S PEAK HOUR

Rule	Normal day peak, 61 orders an hour median · 90% · completed · unfinished · away	Busy day peak, 92 orders an hour median · 90% · completed · unfinished · away	Event day peak, 153 orders an hour median · 90% · completed · unfinished · away
A · fixed responsibilities	1:07 · 1:34 · 58 · 1 · 0	1:14 · 2:09 · 88 · 2 · 0	4:10 · 6:28 · 134 · 11 · 7
B · flexible assistance	1:07 · 1:32 · 58 · 1 · 0	1:13 · 1:54 · 88 · 2 · 0	2:04 · 4:18 · 144 · 7 · 1

SLIDE 10 · FIVE SETUPS: THE FEWEST PEOPLE THAT KEEP WALK-AWAYS WITHIN 2%

Setup	Normal day people · away · wait · S$/day	Busy day people · away · wait · S$/day	Event day people · away · wait · S$/day
Today's split (our scenario)	1 · 0.9% · 2:08 · S$372	2 · 0.0% · 1:14 · S$402	2 · 1.2% · 2:38 · S$402
Machine-made drink (scenario)	1 · 0.0% · 1:14 · S$435	1 · 0.0% · 1:28 · S$435	2 · 0.0% · 1:06 · S$465
All by hand	2 · 0.3% · 1:16 · S$361	3 · 0.0% · 1:13 · S$392	4 · 0.0% · 1:24 · S$422
All robots	– · 0.0% · 0:52 · S$562	– · 0.0% · 0:52 · S$562	– · 0.0% · 0:58 · S$562
One machine (tea and milk)	2 · 0.0% · 1:10 · S$305	2 · 0.1% · 1:36 · S$305	3 · 0.2% · 1:44 · S$335

THE FORECAST'S EFFECT AT THE COUNTER: NORMAL-DAY ROSTER → FORECAST ROSTER, SAME EQUIPMENT AND CUSTOMERS

Setup, bucket brigade	Normal day crew · away · S$/day	Busy day crew · away · S$/day	Event day crew · away · S$/day
Today's split (our scenario)	1→1 · 0.9%→0.9% · S$372→S$372	1→2 · 20.8%→0.0% · S$372→S$402	1→2 · 53.8%→1.2% · S$372→S$402
Machine-made drink (scenario)	1→1 · 0.0%→0.0% · S$435→S$435	1→1 · 0.0%→0.0% · S$435→S$435	1→1 · 5.4%→5.4% · S$435→S$435
All by hand	2→2 · 0.3%→0.3% · S$361→S$361	2→3 · 5.8%→0.0% · S$361→S$392	2→4 · 40.1%→0.0% · S$361→S$422

Peak hour, mean of the seeded runs; cost is the day's roster and machines. Illustrative assumptions, not Chagee data.

SPEAKER NOTES · COUNTER COMPARISONS, EVERY DAY TYPE

Backup for the Serve chapter: the forecast-error sensitivity (slide 08), the three sharing rules (bucket-brigade appendix), the five setups (slide 10) and the forecast's effect at the counter, for every day type.

 TBC · DRAFT, TO BE CONFIRMED

Four modelled decisions by owner: where to open for the expansion manager; what to automate, how many people and how to share for the operations manager

	DECISION	WHY, FROM THE MODEL	REVISIT WHEN
1 **Where to open** expansion manager · slide 12 · DBSCAN + assumed demand model	**Open where app searches cluster, not beside an existing outlet.**	**+731** against **+216** cups a day for the same new outlet. Synthetic app-search data: this shows the method, not a finding about real outlets.	Rerun on Chagee's app logs, if searches and location are recorded, before any lease.
2 **What to automate** operations manager · slide 10 · physical AI	**In our model, one tea-and-milk machine is the cheapest on every day type; pilot it against today's split (our scenario).**	**S$305–335** a day against S$372–402. The two cost the same at about **S$905 a month per machine unit** (1 unit against 3), with wage, volume, station speeds and crews held fixed; we assume S$1,909. Amacha's S$10,000 is a one-off price; without machine life, upkeep and financing we cannot turn it into a monthly cost.	When Chagee's real machine cost is known.
3 **How many people** operations manager · slide 08 and the benefit slide · forecast + economic rule	**Economic rule: add a person on event days, not on busy days.**	With the one machine: peak crew 2 → **3** on event days; 2 on busy days, where an extra person costs more than the orders it keeps.	When the forecast peak moves.
4 **How to share** operations manager · bucket-brigade appendix	**Test a bucket brigade against fixed posts for the hand steps.**	In the recommended setup (one machine, 3 people, event day): **0.2%** walk away with a brigade, 0.4% with well-assigned fixed posts, 22% with badly assigned ones; median wait 1:44 against 2:18.	Fixed posts once there is spare staff.

- **Pilot in 3 outlets**

Shadow	Roster	Machine
forecast only	economic rule	one vs today's split

KPIs: cups per labour hour, wait time and walk-aways at the peak. Compare forecast and actual cups weekly.

- **Limits we know**

All figures are illustrative assumptions, not Chagee data. Machine costs and speeds are assumed; breakdowns and breaks are not modelled.

- **To confirm with Chagee**

The real machine cost per unit, whether outlets log wait times and walk-aways, and how today's local staffing calls compare with the forecast.

ASSUMED BASELINE illustrative — One machine and a flat crew of 2 every day: **23%** of event-day peak orders walk away.

PROPOSED SCENARIO in our model — The forecast and the economic rule add a person on event days only (2 → 3): **0.2%** walk away; busy days stay at 2 (0.1%).

SPEAKER NOTES · THE DECISIONS IN DETAIL

TBC: this slide is a draft and still to be confirmed by the team; present it as work in progress.

The detail behind two of the three owners' decisions: four modelled decisions, one per simulation, all on our assumptions. The CRM manager's decision is in the appendix. One, for the expansion manager, where to open: DBSCAN clusters app searches and our assumed demand model scores the sites. The searches are synthetic, so this shows the method; rerun it on Chagee's app logs, if searches and location are recorded, before any lease.

Two to four belong to the outlet GM or operations manager. Two, what to automate: in our model one tea-and-milk machine is the cheapest on every day type, so pilot it against today's split, which is our scenario, not Chagee's disclosed setup. The two cost the same at about S$905 a month per machine unit, comparing one unit against three, with the wage, volume and crews held fixed. We assume S$1,909. We cannot turn the one-off price one small shop paid into a monthly cost without machine life, upkeep and financing, so Chagee's real cost decides this.

Three, how many people: the economic rule adds a person on event days but not on busy days. Four, how to share: test a bucket brigade against fixed posts; in the recommended setup the brigade does a little better than well-assigned fixed posts and much better than badly assigned ones. Then a pilot in 3 outlets: shadow the forecast, roster by the economic rule, and run one machine against today's split, judged on cups per labour hour, wait time and walk-aways. About one minute.

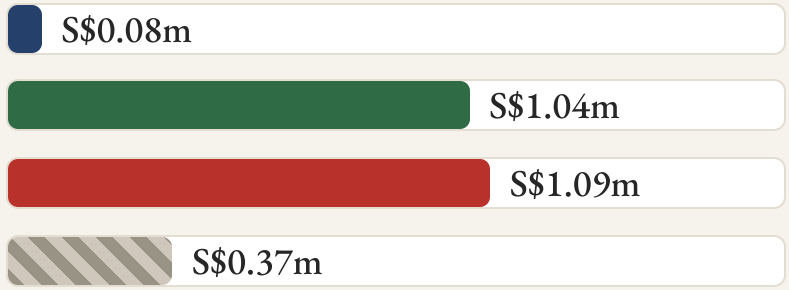
 TBC · DRAFT, TO BE CONFIRMED

In our model, about S$1.09m a year across 44 outlets against an illustrative baseline, almost all from one machine's labour saving

Modelled avoided costs on stated assumptions, not demonstrated savings and not savings against Chagee's actual operations, where the GM and teams staff on local expertise. The pilot compares one machine with today's split (our scenario).

PROFIT A YEAR, ALL 44 OUTLETS · ECONOMIC RULE THROUGHOUT

AGAINST THE ILLUSTRATIVE ALL-HAND BASELINE (FLAT NORMAL-DAY ROSTER)



PILOT COMPARISON: THE PROPOSED SCENARIO AGAINST TODAY'S SPLIT (OUR SCENARIO), BOTH ON THE ECONOMIC RULE



Profit = orders kept × price × margin, plus labour saved, minus machine cost. Calendar: 104 busy days (weekends) and 11 event days (public holidays) a year, an illustrative definition. Baseline: all by hand, the normal-day crew every day. Peak hour only, so orders kept are a floor. Every figure, range and both crew rules: appendix.

PHYSICAL AI · ONE TEA-AND-MILK MACHINE

Most of the benefit: labour

Lets the roster drop from 2 people to 1 in the off-peak hours: **S$1.04m** a year after the machine's cost.

FORECAST + ECONOMIC RULE

Adds event-day orders on top of the machine

Adds a person on event days only (2 → 3), keeping about 35 more orders per outlet that day: 390 a year, **S$0.05m** across Singapore. On its own, all by hand: S$0.08m.

READ WITH CARE

The big number rests on one assumption

The minimum crew, 2 by hand and 1 with machines, is ours. Without it, S$1.09m falls to **S$0.37m**. The pilot arm beats today's split by S$1.22m: more than the headline, because today's split is itself S$0.14m below the baseline.

ASSUMED BASELINE
illustrative

Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

PROPOSED SCENARIO
in our model

Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

SPEAKER NOTES · THE SIZE OF THE BENEFIT

TBC: this slide is a draft and still to be confirmed by the team; present it as work in progress.

One number and where it comes from, all in our model. Against an illustrative baseline, an all-by-hand counter with the normal-day crew every day, the proposed scenario gives about S$1.09m a year across 44 outlets. That baseline is ours, not Chagee's operations, so these are not savings Chagee would see; they are modelled avoided costs on stated assumptions. Every figure uses the economic rule, the rule the playbook recommends.

Read the bars. Almost all of it is the one tea-and-milk machine, which lets the roster drop a person off-peak. Rostering to the forecast by the economic rule adds a person on event days only and keeps about 35 more orders per outlet that day, S$0.05m a year across Singapore. The last bar is the pilot comparison: the proposed scenario against today's split, our scenario, both on the economic rule, a difference of S$1.22m.

Say the caveat out loud: the machine's saving comes from our minimum-crew assumption; without it, the total falls to about S$0.37m. Ranges and both crew rules are in the appendix. About one minute.

Per outlet, about 61 orders kept on an event day and 5 on a busy day; on an illustrative calendar, S$7,539 of revenue a year

Modelled avoided costs on stated assumptions, not demonstrated savings, against an illustrative baseline, not Chagee's operations. Economic rule for the headline figures. Peak hour only; conservative. Prices to our knowledge, based on public sources.

Profit a year, 44 outlets: **S$1.09m** = counter automation, physical AI **S$1.04m** (depends on the minimum-crew rule) + forecast roster, predictive AI **S$0.05m** | Without / with the minimum-crew rule: **S$0.37m to S$1.09m**

ONE OUTLET AT 556 CUPS A DAY · BOTH LEVERS, ONE RUN

EVENT DAY
61 orders kept
Revenue S$358 (× S$5.90) · profit S$241
Counter: labour saved − machine S$27

BUSY DAY
5.4 orders kept
Revenue S$32 (× S$5.90) · profit S$76
Counter: labour saved − machine S$57

NORMAL DAY
0.2 orders kept
Revenue S$1 (× S$5.90) · profit S$58
Counter: labour saved − machine S$57

A YEAR	PER OUTLET	44 OUTLETS, IF ALL LOOKED LIKE THIS MODEL
Orders kept peak hour, busy and event days	**1,278** range −121 to 1,452	**56,221**
Revenue orders × price	**S$7,539** range −S$832 to S$10,015	**S$331,704**
Profit margin 50%–70%: −S$754 to +S$754	S$24,660	S$1,085,027
Labour saving, counter labour saved − machine cost; shown on its own	S$20,136	S$886,005

Revenue and labour are never added: revenue is orders × price; profit is orders × price × margin, plus labour saved, minus machine cost.

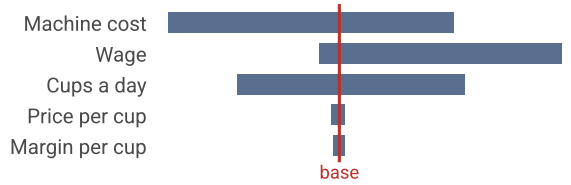
Calendar: 104 busy days (weekend days) and 11 event days (Singapore public holidays) a year: an illustrative definition, to confirm with Chagee; sliders on screen.

ORDERS · PROFIT A YEAR	WALK-AWAY RULE	ECONOMIC RULE
Forecast crew	1,242 · S$603	670 · S$1,711
Machine	888 · S$23,611	888 · S$23,611
Both, one run	**1,278 · S$24,660**	**1,278 · S$24,660**
Both, today's split counter −S$305/month	1,124 · −S$3,142	1,124 · −S$3,142

Walk-away rule: fewest people keeping peak-hour walk-aways within 2% (the forecast crew at 556 cups a day). Economic rule: add a person only when orders kept × price × margin beat the extra labour. "Both" is one run, not a sum: the levers keep many of the same orders, and with the machine the forecast crew needs fewer extra hours (−S$662 profit against the sum). Headline figures use the economic rule.

Against what
A flat normal-day roster and an all-by-hand counter. Chagee deploys people on local expertise (Chagee source, anonymous), so this baseline is ours, not Chagee's.

All inputs at their worst/best end together
Profit a year, one outlet: low **−S$15,660** · base **S$24,660** · high **S$101,166**



Fragile: the minimum crew
The counter lever rests on the model's minimum crew off-peak: 2 by hand, 1 with machines. Profit a year per outlet: S$24,660 with it; S$8,395 with 1 either way.

Not sized
Delivery levers (third-party; channel margins not disclosed), membership and CRM, waste (no inventory model in this deck), and the new-store site forecast (shown as cups added only).

ASSUMED BASELINE illustrative
Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

PROPOSED SCENARIO in our model
Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

SPEAKER NOTES · THE BENEFIT IN DETAIL

Backup for the benefit slide. This answers part c of the brief: how big is the benefit, for Singapore only. Against what: a flat normal-day roster and an all-by-hand counter. Chagee deploys people on local expertise, the GM and outlet teams making the calls, so this baseline is ours, not Chagee's practice.

Lead with orders. On an event day the model keeps about 61 orders per outlet in the peak hour that would otherwise walk away, about S$358 of revenue; on a busy day about 5. Over a year it depends on how many such days there are: our illustrative calendar is 104 busy days, the weekends, and 11 event days, the public holidays. That is an input, not a measurement.

Profit is the second line, using a margin we assume because Chagee does not disclose one. The labour saving is shown separately and never added to revenue. The two levers are run together, not added, because they share the same labour hours. The strip at the top splits the profit: most is counter automation, physical AI, the machine on its own; the forecast, predictive AI, adds a small share on top.

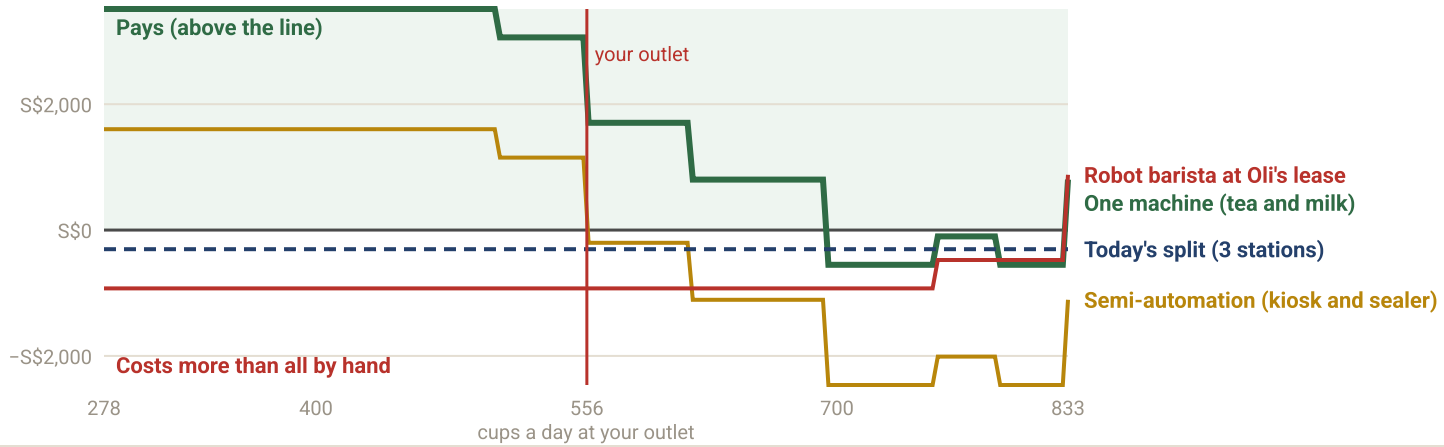
Two crew rules for rostering to the forecast. The walk-away rule adds people until peak-hour walk-aways are within 2%; at our default volume that is exactly the forecast crew. The economic rule adds a person only when the orders kept are worth more than the extra hours. They differ on busy days by hand, where the economic rule adds no one.

Be honest about fragility: most of the profit is the counter's labour saving, and it comes from the model's minimum crew, 2 by hand and 1 with machines. Drop that rule and the profit falls to about S$8,395 a year per outlet. And rostering to the forecast pays on event days, not on busy days, where the extra hours cost more than the orders won back. These are modelled avoided costs on stated assumptions, not demonstrated savings. About one minute.

A machine pays only when it removes a person-shift for less than the shift costs

To our knowledge, based on public sources; vendor list prices, not cost to implement. Chagee's equipment cost is not public; its Singapore relaunch used semi-automated equipment (Vulcan Post).

NET GAIN A MONTH AGAINST ALL BY HAND · ONE OUTLET



Shown at 556 cups a day, wage S$15.06 an hour, S$1,909 a month per machine unit (sliders on screen).

	1 Semi-automation	2 Beverage platform	3 Milk-tea machine	4 Robot arm	5 Robot barista
	Gong Cha Singapore	Gong cha Super Wu	Amacha, one shop	Richtech ADAM	Oli; coffee-first
	Not disclosed	*Not disclosed*	S$10,000 to buy	*Not disclosed, rental only*	US$4,990/mo lease
	PRESS	COMPANY CLAIM	PRESS	VENDOR PAGE	VENDOR PAGE

The model's S$1,909 a month is per station unit (our reading: a semi-automated station). Semi-automation and the beverage platform are priced at it because their real prices are not disclosed; robot prices are per setup. Full table and sources in the appendix.

START HERE

One machine on the tea and milk

+S$1,706 a month vs all by hand

Saves 8 person-hours a day (S$3,614 a month) for one S$1,909 unit. Simple payback, if it lets the roster drop one person-shift: 2.8 months on Amacha's S$10,000.

WATCH

Today's 3-station split

−S$305 a month vs all by hand

Saves 12 person-hours a day (S$5,422) but pays for 3 units (S$5,726).

WAIT FOR VOLUME

Robots: arm or barista

757+ cups a day before they save more labour than today's split (labour only, not a cost break-even)

At 556 cups, Oli's lease (S$6,350) is S$928 a month worse; a robot setup would need to cost under S$5,422 a month. The arm's price is not disclosed.

Model assumption that drives all three: minimum crew 2 by hand, 1 with machines, off-peak. Station speeds are the model's, not vendor claims. Rosters step in whole people, so results flip with volume and wage.

WITHOUT AI
our assumed baseline

All by hand at 556 cups a day: **24 person-hours** a day, **S$10,843** a month of labour.

WITH AI
our proposal

One tea-and-milk machine: **16 person-hours** a day, **+S$1,706** a month after its unit cost. A robot setup (four stations on machines) starts to save more labour than today's split at about 757 cups a day; labour only, not a cost break-even.

SPEAKER NOTES · MACHINE LADDER

Backup for slide 10, for questions on which machine to buy. This is physical AI: automation and machines at the counter. The predictive AI is the forecast that sets the roster. One rule explains the whole slide: a machine pays only when it removes a person-shift, and costs less than that shift.

The chart shows the net gain a month against an all-by-hand counter as volume changes; above the line it pays. One tea-and-milk machine lets the roster drop from 2 people to 1 in 8 off-peak hours, about one person-shift a day, for one station unit: it pays across most of the range. Today's split saves 12 person-hours but pays for 3 units, so it ends up a little dearer than all by hand. A robot setup, four stations on machines, saves no more labour than today's split until about 757 cups a day. That is where extra labour saving begins, not a cost break-even; at Oli's lease it only pays at the top of our range.

Three things to say out loud. The minimum crew, 2 by hand and 1 with machines, is our model's assumption and drives this result. Every option runs at the model's station speeds, not vendor claims. Payback is simple payback, if the machine lets the roster drop one person-shift, with no installation, maintenance or financing. Chagee's own equipment cost is not public. About one minute.

Takeaway: At 556 cups a day: start with one machine (about +S$1,706 a month); today's 3-station split costs S$305 a month more than all by hand; robots save no more labour than today's split until about 757 cups a day.

Every rung, every price as published, and where it comes from

To our knowledge, based on public sources. Physical AI: automation and machines. A gap is "not disclosed", never a guess.

RUNG	EXAMPLE	AUTOMATES · STEPS ON MACHINES (OUR MAPPING)	VENDOR LIST PRICE	CLAIMS (NOT MODEL INPUTS)	RELIABILITY	SOURCE
1 **Semi-automation**	Gong Cha Singapore	Sugar dispensers, cup sealers, self-order kiosks. **Order and tag, Shake and seal**. Kiosk = order and tag, sealer = shake and seal. Sugar dispensing sits inside tea and milk, so it is not modelled.	*Not disclosed*	Over 20% productivity from self-ordering; 5% manpower saving (no baseline given)	PRESS	https://www.ntuc.org.sg/uportal/news/Technology-Brewing-Up-the-Bubble-Tea-Business/
2 **Beverage platform**	Gong cha Super Wu (Feb 2026)	Dispenses tea, sweeteners and syrups; brewing and shaking stay manual. **Tea and milk**	*Not disclosed*	Up to 65% higher peak productivity	COMPANY CLAIM	https://marketech-apac.com/gong-cha-launches-gong-cha-2-0-to-modernise-global-operations-and-store-design/
3 **Single automated milk-tea machine**	Amacha (one small shop, owner interview)	Makes the milk tea itself. **Tea and milk**	S$10,000 to buy	The shop's total opening cost was S$209,000 (not a machine cost)	PRESS	https://www.malaymail.com/news/life/2026/10/05/opening-a-milk-tea-shop-in-singapore-can-rack-up-to-s209000-even-before-the-first-cup-is-poured/237715
–	**Our scenario: today's split**	**Order and tag, Tea and milk, Shake and seal**. A scenario of ours (slides 09 and 10), not Chagee's disclosed setup	Model: S$1,909 a month per station unit		OUR ASSUMPTION	assumptions.json
4 **Robot arm that makes boba**	Richtech ADAM	Cup, ice, tea, toppings and sealing by one robot arm. **Cup and ice, Tea and milk, Toppings, Shake and seal**	*Not disclosed*. Rental only; price not published	–	VENDOR PAGE	https://richtechrobotics.com/solutions/adam
5 **Robot barista**	Oli	A whole drink station in one unit. **Cup and ice, Tea and milk, Toppings, Shake and seal**. Coffee-first. No tea drinks listed for Oli or Artly.	US$59,000 (≈ S$75,078) to buy; US$4,990 (≈ S$6,350) a month lease	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Artly		US$68,000–88,000 (≈ S$86,530–111,980) to buy; US$2,500–4,000 (≈ S$3,181–5,090) a month lease; US$2,500–4,000 a month software Oli's estimate of a competitor. The figures as given do not say whether the lease replaces or adds to the purchase, so no verdict is computed.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Cafe X		US$250,000 (≈ S$318,125) to buy Oli's estimate of a competitor's price.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
–	**Chagee Singapore**	Chagee's own equipment cost is not public. Its Singapore relaunch used semi-automated equipment.	*Not public*	–	PRESS	https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/

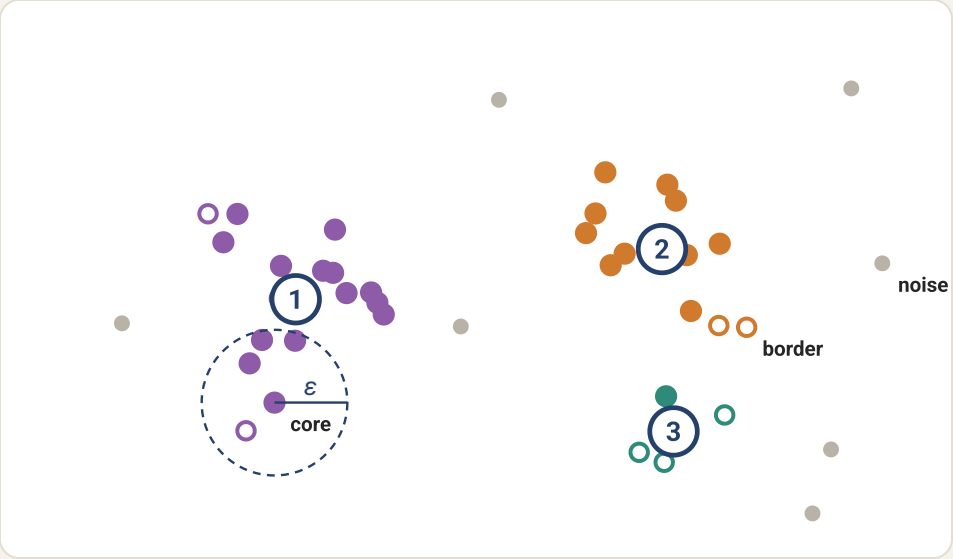
Vendor list prices, not cost to implement: installation, training, maintenance, power, space and import are not included; USD converts at S$1.2725 per US$. **Simple payback** = list price ÷ labour saved a month, with no maintenance or financing. **The model's S$1,909 a month** is per station unit: our reading: one semi-automated station unit (for example a dispenser or a cup sealer), amortised with upkeep. Not a robot barista lease; robot prices in machine_tiers.json are per setup. **Vendor speed and throughput claims** are listed, never used: every rung runs at the model's station speeds. **Steps on machines** is our mapping. Artly and Cafe X prices are Oli's estimates of competitors.

SPEAKER NOTES · MACHINE LADDER: TIERS AND SOURCES

Backup for the machine ladder. Every price is as the source publishes it; where a source gives no price we say not disclosed. Claims such as Gong cha's productivity figures are listed for context and are not inputs to our model. Soft copy only.

DBSCAN finds where searches bunch up; a proximity model turns a pin into cups

DBSCAN IN FOUR MOVES · TOY EXAMPLE



1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 12 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

1	**App searches**	Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns	**3,000 dots**
2	**No store nearby**	The nearest outlet is further than the walking distance	**1 km · 46% of dots**
3	**DBSCAN**	Clusters the no-store-nearby dots	**ε 0.6 km · min 15 · 11 clusters**
4	**Pull of a store**	Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance	**L = 1 km**
5	**Who buys where**	A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one)	**not buying 0.3**
6	**Calibration**	Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days)	**556 cups a day**
7	**Score a pin**	Cups for Chagee = what the new store sells − what it moves from nearby outlets	**top: 837 − 106 = +731**
8	**Verdict**	Share of the new store's cups that are new to Chagee	**strong ≥ 60% · weak < 40%**
9	**Opening roster**	Peak-hour cups ÷ cups one person handles, with the safety margin	**one per 83 cups**

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

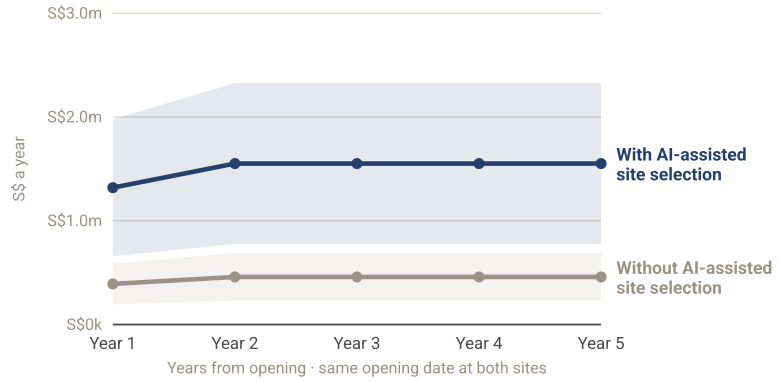
SPEAKER NOTES · HOW THE SITE FORECAST WORKS

Backup for questions on slide 12. Left: DBSCAN in four moves on a toy set. A search is a core point if enough searches sit within the radius; clusters grow from core to core; border points join; the rest is noise. We do not choose the number of clusters.

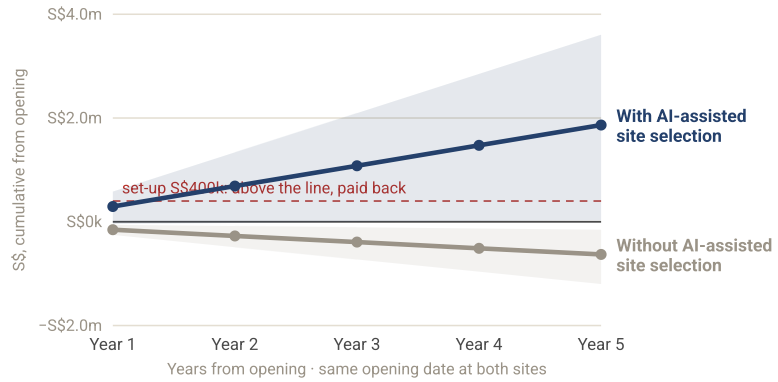
Right: every simulated number, in the order the model uses it. The forecast is only as good as these assumptions; with Chagee's real app logs, the first two rows would be replaced by data and the rest would run as is.

Same new outlet, same opening, same counter: the AI site adds about S$2.5m more operating profit over 5 years, in our model

ANNUAL INCREMENTAL REVENUE TO CHAGEE, S$



CUMULATIVE INCREMENTAL OPERATING PROFIT, S$



THE BRIDGE, FIVE-YEAR TOTALS, BASE SCENARIO

5 YEARS, BASE SCENARIO	WITHOUT AI · +216	WITH AI · +731	DIFFERENCE
New-store sales	S$6.2m	S$8.6m	S$2.4m
− Sales displaced from nearby Chagee outlets	−S$4.0m	−S$1.1m	S$2.9m
= Net additional network sales (incremental revenue)	**S$2.2m**	**S$7.5m**	**S$5.3m**
− New-store ingredients, packaging and delivery commission	−S$3.1m	−S$4.2m	−S$1.2m
− New-store labour (same staffing rule)	−S$325k	−S$412k	−S$87k
− Machines, rent, utilities, maintenance	−S$1.4m	−S$1.4m	S$0k
+ Nearby outlets' variable costs saved (their rent and staff stay)	S$2.0m	S$537k	−S$1.4m
− AI recurring cost (allocated)	S$0k	−S$120k	−S$120k
= Incremental operating profit, before depreciation, interest and tax	**−S$630k**	**S$1.9m**	**S$2.5m**

Illustrative scenarios on our assumptions, not a demonstrated causal gain from AI. DBSCAN finds the search clusters; the assumed demand model estimates the sales. Shaded: the scenario range (our per-outlet volume range), not a confidence interval.

5-YEAR RESULTS · BASE · ILLUSTRATIVE

Revenue uplift	S$5.3m
Operating-profit uplift	S$2.5m
Payback, with AI	1.4 years
Payback, without AI	beyond year 5

Cumulative cash flow after the S$400k set-up (AI site S$460k). AI cost repaid in 2 months. Volume range: S$1.2m to S$3.8m profit uplift.

ASSUMPTIONS · S$ · REUSED UNLESS MARKED NEW

360 days · S$5.90 a cup, held flat · ingredients and packaging 40%
delivery 30% of cups at 30% · wage S$15.06 an hour, same staffing rule
rent S$15,000, utilities S$2,500, maintenance S$800 new a month · same machines
year-one ramp-up 85% new · growth 0.0% new · cost inflation 0.0% new
AI S$60k to build, S$24k a year, 100% on this outlet new · set-up S$400k, outside operating costs

TAKEAWAY

Beside the nearest outlet: **+216** net cups a day → about **S$2.2m** of network sales over 5 years, and **−S$630k** operating profit: the contribution lost at its neighbours outweighs what it makes. With AI-assisted site selection: **+731** net cups a day → about **S$7.5m** and **S$1.9m** after the AI costs: **S$2.5m** more profit than the baseline. Illustrative, not a proven causal gain.

SPEAKER NOTES · THE NEW STORE IN MONEY: FIVE-YEAR P&L

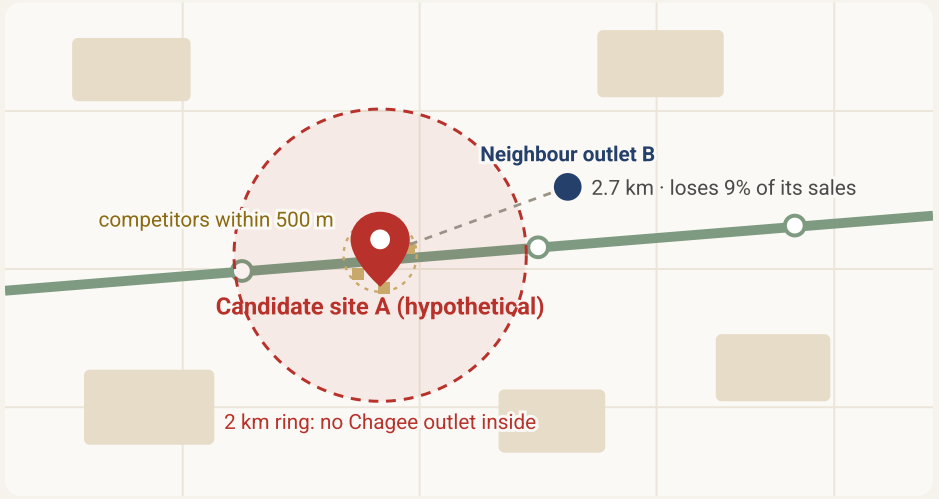
Soft copy and print companion of slide 12's P&L view, at the default inputs and the suggested site. The on-screen view updates with the selected site and the assumption sliders; this page shows the defaults.

Extension: where this scales next. Not included in the S$ benefit.

Four scores for every location

Corporate drops a pin; the model rates it on four things (illustrative). Decision owner here: the expansion manager.

SCHEMATIC MAP · HYPOTHETICAL PINS



Demand score HAND-SET — 84
Residents, workers and transit riders within reach, plus footfall.

Delivery demand score HAND-SET — 78
Orders from the area on third-party delivery apps.
All Chagee delivery goes through third-party apps (anonymous Chagee source, to confirm), so we use area demand, not "served from a farther outlet". Whether the platforms share this data is to confirm.

Competition score HAND-SET — 61
Bubble-tea and cafe outlets within 500 m; dense competition lowers the score, a few prove demand.

Cannibalisation factor COMPUTED · SLIDE 12 ENGINE — −13
Site A: 13% of its cups would move from existing Chagee outlets; neighbour outlet B loses 9%. Deducted so the result is net to Chagee.

Site score WEIGHTED — 76
Demand 50%, delivery 20%, competition 30% (hand-set weights).

ATTRIBUTES THE MODEL TAGS TO EACH PIN residents and workers within 1 km · MRT and bus tap-ins · footfall · delivery orders from the area · competing outlets within 500 m · Chagee outlets within 2 km · rent per sq ft · staff needed at peak

WHAT IS AI HERE Scoring a pin from its attributes would be a supervised model trained on outlets' results: a production idea. Here the three scores are set by hand, and the cannibalisation factor is computed by the slide 12 engine on synthetic data. The next slide runs the supervised model.

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it.

SPEAKER NOTES · FOUR SCORES FOR EVERY LOCATION

An extension: where this scales next, and not part of the benefit figure. The decision owner here is the expansion manager. This builds on Jin's site-selection framework.

Step through it. Corporate drops a pin; the model tags its attributes; three scores follow. Demand, delivery demand and competition are set by hand here, and you can drag a bar to test another value: in production, a supervised model would score them. We call the second one delivery demand, not delivery relief, because all Chagee delivery runs through third-party apps, according to an anonymous Chagee source; whether the platforms share this data is to confirm.

The cannibalisation factor is computed by the same engine as slide 12: about 13% of site A's cups would move from existing outlets. Switch to the pin with an outlet inside the ring, or the one beside an outlet, and watch it jump. All synthetic. About thirty-five seconds.

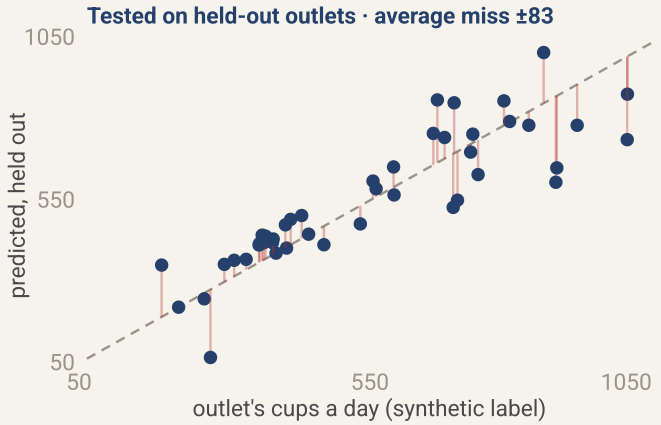
Extension: where this scales next. Not included in the S$ benefit.

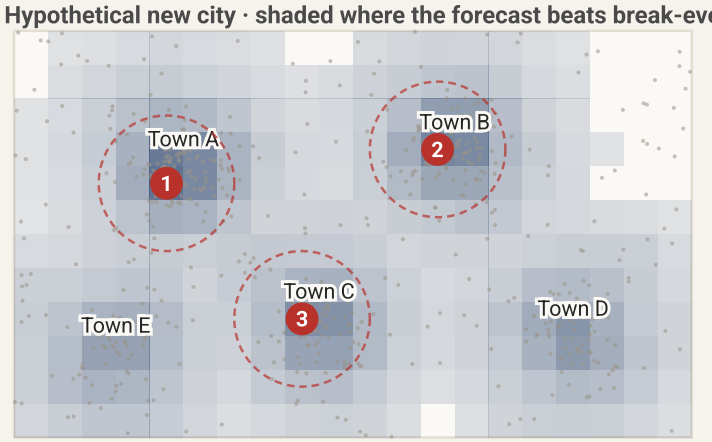
Trained on Singapore's outlets, the model picks where to open in a city where Chagee has no data yet

Supervised learning: keep only the inputs a new city has, test the model on outlets it has not seen, then let it score every spot in a hypothetical city (illustrative, synthetic).

TRAIN ON 44 OUTLETS · TOGGLE THE INPUTS

- ☑ **People within reach** — NEW MARKET: HAS IT
- ☑ **Distance to a town centre** — NEW MARKET: HAS IT
- ☐ Delivery-area demand — NEW MARKET: TO CONFIRM
- ☐ Unserved app searches — NEW MARKET: NONE
- ☐ Chagee outlets nearby — NEW MARKET: NONE
- ☐ Distance to nearest Chagee outlet — NEW MARKET: NONE
- ⬚ Competitors within 500 m — NOT IN OUR DATA
- ⬚ Rent per sq ft — NOT IN OUR DATA





TESTED ON OUTLETS IT HAS NOT SEEN

±83 cups a day, average miss

R^2 0.76 with 2 inputs

All inputs (Singapore)	±41 · R^2 0.95
New-market inputs only	±83 · R^2 0.76
No inputs: guess the average outlet	±197

WHERE TO OPEN IN THE NEW CITY

1. Town A: ≈ **1,090** ± 83 · **above break-even even at 1,007** · **beyond what it learned**
2. Town B: ≈ **959** ± 83 · **above break-even even at 876**
3. Town C: ≈ **887** ± 83 · **above break-even even at 804**

Cups a day, forecast ± average miss. Break-even ≈ 275 at Singapore costs (the pin-to-profit slide), a stand-in. Picks ≥ 2 km apart.

WHAT IS AI HERE Supervised learning (ridge regression) from site attributes to cups a day. The labels are the slide 12 engine's synthetic outlets, so the fit flatters; Chagee's real opening history would fit worse. Using Singapore to score another market assumes its customers behave alike: check against the first outlets' actuals.

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it.

SPEAKER NOTES · THE SUPERVISED MODEL

This is the model behind step three of the framework, actually running. It learns cups a day from a site's attributes, trained on Singapore's 44 outlets. The labels are synthetic, from the slide 12 engine, so read it as the method, not a forecast.

We test it fairly: each outlet is predicted by a model that never saw it. With every input, it misses by about 41 cups a day on average. Now the proxy for a new market: there is no Chagee app history and no outlet network yet, so switch those inputs off. With only population and distance to a town centre, the error grows to about 83 cups a day, still better than guessing the average outlet, about 197.

Then the point of it: where to open in a place with no local data. The right-hand map is a hypothetical city, generated at the same density as our Singapore data. With only the inputs that city has, the model scores every spot and picks the best sites at least 2 km apart, so they do not take sales from each other. Each pick shows its forecast with the error band, and whether it clears break-even at Singapore costs, which stand in until local costs are known. Picks beyond what the model saw in Singapore are flagged. So the model can travel, but only on inputs the new market has, with wide ranges: pilot the top site first and recheck the model against its actuals. That is the continual-learning step on the next slide. About fifty seconds.

Extension: where this scales next. Not included in the S$ benefit.

From pin to net profit

Six steps from a pin to a go / no-go, on profit net to Chagee overall (illustrative). Decision owner here: the expansion manager.

1 Pin a location
Corporate clicks a point on the map.
NOT AI

2 Tag attributes
People, transit, footfall, delivery, competitors, outlets nearby, rent.
DATA

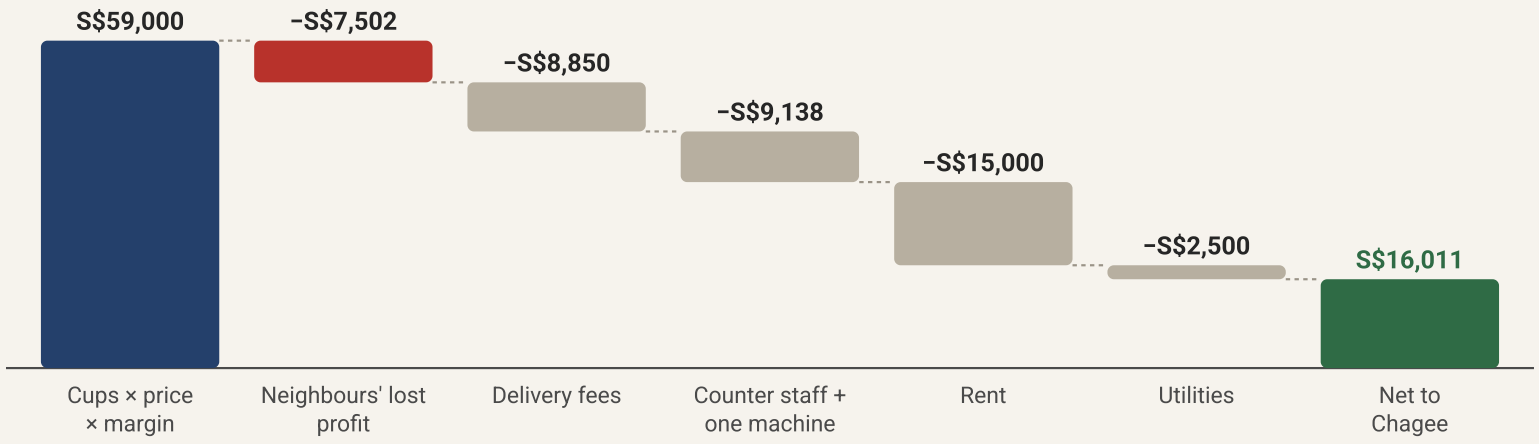
3 Four scores
Demand, delivery, competition, cannibalisation (previous slide).
AI · SUPERVISED MODEL

4 Find look-alikes
Anonymised outlets (Outlet 12, Outlet 27): their cups a day, what neighbours lost.
AI · SIMILARITY SEARCH

5 Forecast net profit
Cups a day, running cost, minus the profit neighbours lose.
COMPUTED BELOW

6 Decide
Open, pilot or skip. Actuals feed back: continual learning.
AI · CONTINUAL LEARNING

NET PROFIT TO CHAGEE, A MONTH · SITE A, TOP APP-SEARCH CLUSTER



At 278–833 cups a day: **−S$3,506** to **S$34,624** a month. Neighbour outlet B, 2.7 km away, loses 9% of its sales. Delivery fees: 30% of cups × 30% commission. Counter: the one-machine setup, normal-day roster (counter model).

STEP 6 · RECOMMENDATION

Pilot

positive at 556 cups a day, negative at 278, the low end: test it before a full lease.

Site score **76 / 100 (four scores)**
Payback **≈ 25 months on S$400,000**

A strong-demand site can still fail if rent or staffing is too high, or if it mostly takes sales from our own neighbours.

In practice this needs Chagee's opening history. With about 44 outlets in Singapore the model must stay simple (few attributes, ranges not point values).

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it. Cups a day, price and margin are the benefit slide's; rent, delivery, utilities and set-up cost are assumptions, not Chagee data.

SPEAKER NOTES · FROM PIN TO NET PROFIT

Six steps from a pin to a go or no-go. Steps three, four and six are where the AI would sit in production: a supervised model for the scores, a similarity search for look-alike outlets, and continual learning as each new outlet reports actuals. On this slide only step five is computed.

The waterfall starts from cups a day times price times margin, the same inputs as the benefit slide: 556 cups a day. Take off the profit neighbours lose, delivery fees, the counter, rent and utilities: about S$16,011 a month net to Chagee, a payback of about 25 months. At the low end of the cups range it loses money, so the call is: Pilot. Move rent, delivery or set-up cost and watch the call change; these are our assumptions, not Chagee data.

The honest point: switch to the pin with an outlet 1.7 km away. It takes 44% of its cups from our own outlets and comes out at −S$2,697 a month, a skip. About forty-five seconds.

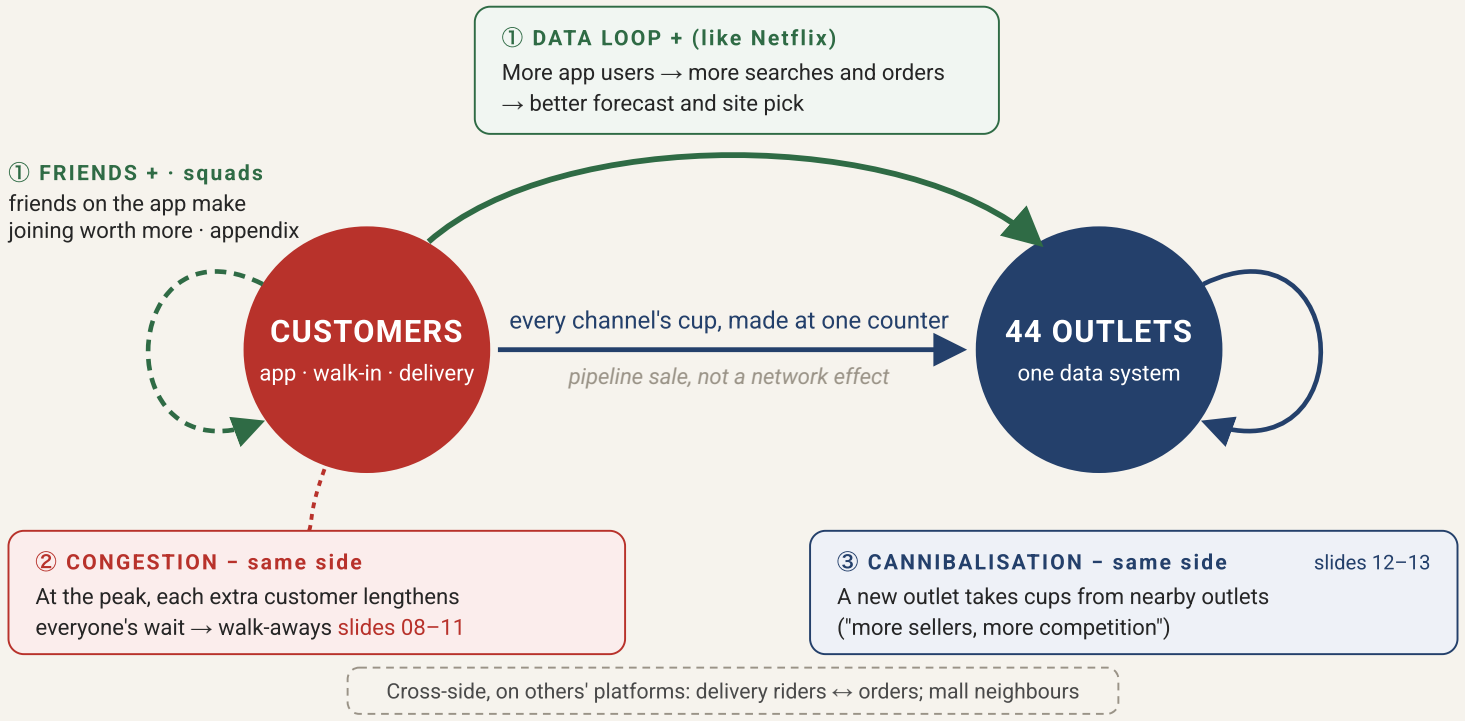
Later: reuse in new markets via transfer learning, out of scope.

Course link · cross-side vs same-side network effects (eBay)

Chagee's network effects sit on the customer side: we grow the friend and data loops, then cap the crowding they bring

Our reading: Chagee sells its own tea, so there is no seller side to match. Part one grows the positive loops; parts two and three cap the crowding.

WHERE CHAGEE SITS ON THE DIAGRAM · CLICK A LOOP



① GROW · PART ONE · APPENDIX

Make friends worth more, feed the data loop

Squads: friends taking part raise the features' value (not AI). **Why Chagee:** app-first and member-led: **257 million** members group-wide; repeat members place **more than 78%** of orders (earnings call, Aug 2026; SG to confirm).

② CAP · PART TWO · SLIDES 08 TO 11

Absorb the peak crowd

Forecast crew plus a bucket brigade: event day walk-aways **54%** → **1.2%** of peak-hour orders.

③ CAP · PART THREE · SLIDES 12–13

Count cups added, not sold

Of 837 cups at the new site, **106** (13%) move from nearby outlets. The score subtracts them.

NOT OUR LEVER

Delivery apps and malls own the cross-side effects; their riders still count in our peak.

Takeaway | Grow the friend and data loops, then make sure the counter and the outlet map can absorb the extra customers.

SPEAKER NOTES · NETWORK EFFECTS ACROSS THE THREE PARTS

This is how the three parts use network effects. A lens from the course: on eBay, buyers and sellers attract each other, a cross-side effect. Chagee is not that, in our reading: it sells its own tea, so customers to outlets is a pipeline sale. Its network effects sit on the customer side.

Two we grow, in part one. Friends: with squads, each friend on the app makes joining worth more to the others; the members slides in the appendix model that with an effort score and test whether it pays. That part is analytics and product design, not AI. And the data loop: more app use gives more search and order data, so the forecast and the site pick get sharper. That is why we chose Chagee: on its second-quarter 2026 call it reported 257 million registered members, and members with two or more purchases placed more than 78% of orders. Group-wide figures; Singapore's app share is to confirm.

Two we cap. Part two, congestion: at the peak every extra customer lengthens everyone's wait; a normal-day roster on an event day loses about 54% of peak-hour orders, and a forecast crew with a bucket brigade brings it to about 1.2%. Part three, cannibalisation: a new outlet takes cups from its neighbours; of 837 cups at the app-search site, 106 move from nearby outlets.

Cross-side effects exist, but on others' platforms, delivery apps and malls: not our lever. And we do not carry any modelled membership uplift into the counter or outlet numbers; each part stands on its own assumptions. About fifty seconds.

Where every number comes from

FACTS QUOTED

Outlets in Singapore: 44. Store count used on the storyboard deck. The doc also cites 46 (VnExpress, earlier); the later directory count is used. *[L-sg-directory]*

Global context, group-wide, not Singapore: franchise share of FY2025 net revenue: 0.88. Secondary source; flagged wherever shown. GROUP-WIDE: global context only, never applied to Singapore (Singapore is company-owned, I-sg-owned). *[L-koalagains]*

Global context, group-wide, not Singapore: company-owned share of FY2025 net revenue: 0.12. Secondary source; flagged wherever shown. GROUP-WIDE: global context only, never applied to Singapore (Singapore is company-owned, I-sg-owned). *[L-koalagains]*

Registered members, group-wide, end of June 2026: 257000000. Company statement on the Q2 2026 earnings call (COO, interpreted). Group-wide, not Singapore. Membership is on Chagee's mobile mini program (L-chagee-q1-2025). *[L-chagee-q2-2026-call]*

Orders from members with two or more purchases, group-wide (more than): 0.78. Company statement on the Q2 2026 earnings call. A floor ('more than 78%'). Group-wide, not Singapore. *[L-chagee-q2-2026-call]*

SOURCES

L-sg-directory Chagee Singapore store directory, mapped by the team (44 locations, checked 15 Oct 2026); Mothership, Sep 2026, https://mothership.sg/2026/09/chagee-more-heartland-stores-singapore-eugene-lee/ gives the same count. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

I-sg-owned PROJECT OWNER CONFIRMATION (23 Oct 2026): Chagee Singapore operates a company-owned model, not a franchise model. Confirmed by the project owner; not from the anonymous Chagee contact and not an external citation. Used on slide 01 (headline, step 1, card, notes). Supersedes the earlier 'ownership not claimed' framing. It does not change the group-wide revenue shares, which stay global context only.

L-koalagains KoalaGains, 'CHA business and moat', https://koalagains.com/stocks/NASDAQ/CHA/business-and-moat (SECONDARY - flag when used): FY2025 net revenue about 88% from the franchise (master franchise) model and about 12% from company-owned stores, the latter growing fastest. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-yfchuhai YF Chuhai, https://www.yfchuhai.com/article/9751.html: Chagee's leaf supply chain is in China; it sources only two ingredients locally, fresh milk and fruit. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-sec20f Chagee Holdings Form 20-F for FY2025, SEC EDGAR, https://www.sec.gov/Archives/edgar/data/2013649/000110465926050766/cha-20251231x20f.htm: separate centralized and regional warehouses to lower storage and logistics costs. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-unsw UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)

I-chagee-contact CHAGEE SOURCE, ANONYMOUS (Oct 2026): a Chagee source who asked not to be named; do not identify them. Four points only: whether Singapore outlets are owned, a joint venture or franchised does not matter for this work, because the data sits in one system; Singapore outlets run under one company and one data system; all delivery is via third-party apps; people are deployed on local expertise: the GM and outlet teams make the staffing calls. Ownership itself is not attributed to this source: see I-sg-owned (project owner confirmation).

I-sg-groundfloor INFORMAL - quote in the team's ideation doc (Chagee Singapore's Lee: 'Whenever you see a Chagee, it's always on the first floor'; kiosk-style outlets in basements and MRT stations at entry in 2019). The article is not named in the doc; confirm the source before quoting.

I-team-app-first INFORMAL - team observation in the ideation doc: Chagee Singapore sells mainly through its app, with minimal order kiosks in outlets. Confirm before citing. SUPERSEDED on slide 01 (22 Oct 2026) by P-appstore, P-vulcanpost-comeback and L-sec20f-data; the app's share of orders stays to confirm.

L-ester1996 Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231. The DBSCAN algorithm used on slide 14.

P-honeycombers PRESS - in-store sightings, indicative. Honeycombers (Feb 2026), https://thehoneycombers.com/singapore/china-bubble-tea-brand-singapore/ : Chagee price sightings in Singapore; not a current full menu.

P-eatbook PRESS - in-store sightings, indicative. Eatbook (Aug 2024), https://eatbook.sg/?p=168476 : Chagee price sightings in Singapore; not a current full menu, and older than the deck.

I-team-expansion INFORMAL - team research (Oct 2026): Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential. Singapore only. Confirm before citing.

I-team-site-framework TEAM WORK (Oct 2026): Jin's site-selection framework, two team slides ('Four Scores for Every Location', 'From Pin to Profit Forecast'): four scores per pin, six steps from a pin to a go / no-go on profit net to Chagee. Used for the appendix pin-to-profit page (earlier extension). Its example values and outlet names are not used; ours are illustrative.

L-chagee-q2-2026-call Chagee (CHA) Q2 2026 Earnings Call, full transcript, TradingKey, 29 Aug 2026 (local copy: 'Chagee (CHA) Q2 2026 Earnings Call_ Profit Margin Improves as Overseas GMV Doubles.pdf', supplied by the team). COO's remarks, interpreted: 'As of the end of June, our total registered members reached 257 million. The repurchase rate among active members remain above 43%, while members who made 2 or more purchases accounted for more than 78% of total orders.' Group-wide, not Singapore. The page's AI summary is not used; the quote is from the transcript.

L-chagee-q1-2025 Chagee Holdings, 'Chagee Announces First Quarter 2025 Unaudited Financial Results', GlobeNewswire, 30 May 2025 (local copy supplied by the team; the team's link https://investor.chagee.com/node/6736/pdf blocks automated access, so not checked by us). 'Chagee's Mobile Mini Program had a total of 192.4 million registered members as of March 31, 2025'; active members = registered members who placed an order at least once in a period. Shows that membership sits on Chagee's ordering app. Group-wide.

P-appstore CHAGEE app, App Store listing (team research, Oct 2026; no link supplied, not checked by us): 'Order and pay digitally', 'Order for self-pickup ahead of time', 'Locate our outlets and their operating hours'. The listing seen was the Saudi storefront; its privacy-policy URL includes 'SGMY' (our reading: Singapore and Malaysia, an inference). Privacy label: Precise Location under 'Data Not Linked to You'; search history not listed.

P-vulcanpost-comeback PRESS - Vulcan Post, Jayme Teo, 2 Oct 2024, https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/ : after the reopening, with some customers 'waiting for as long as an hour to receive their orders', 'the brand rolled out in-app ordering and delivery and constantly upgraded the ordering process.' Its 'kiosks' are a store format ('pivoting from their takeaway kiosks for comfortable dine-in, sit-down tea bars'), not ordering kiosks. Quotes checked by us, 22 Oct 2026.

L-sec20f-data Chagee Holdings Form 20-F for FY2025 (same filing as L-sec20f), risk factors: 'Our business generates and processes personal, transactional and other data.' and 'we may not be able to use our membership program to deepen our understanding of our target consumers.' No share of orders by app or mini program, and nothing on search or location data, in the parts we read. Quotes checked by us, 22 Oct 2026.

L-sec20f-member Chagee Holdings Form 20-F for FY2025 (same filing as L-sec20f), Item 3.D risk factor 'We may not be successful in expanding our membership and our ability to take advantage of our membership program may be limited.' Both quoted sentences checked word for word by us against the SEC EDGAR filing, 22 Oct 2026.

C-topic8 OPIM639 course slides, Topic 8 (slides08 Service Platforms), slides 30-36: the economy with network effects, r(x) = 1 - x, f(z) = z, equilibria z' and z". Slide 34 quoted word for word, checked by us against the course file, 22 Oct 2026.

SPEAKER NOTES · SOURCES AND ASSUMPTIONS

Soft copy only.

Every Australian input is a scenario assumption

STATUS

Australia worked example, kept separate from every other section: nothing here feeds the main story (slides 01 to 16) or the benefit and nothing in them is changed. Every Australian input is a scenario assumption; no Australian data has been retrieved for this deck. Archetype areas stand in for real suburbs until Census and market data are pulled. Market status: no local Chagee operating history (assumed for this worked example).

CANDIDATE AREAS (ARCHETYPES, SCENARIO ASSUMPTIONS)

A · CBD office core: residents within 1 km 12, workers within 1 km 90, tertiary students within 1 km 15, household income index 115, transit access 0.95, competitors within 500 m 9, rent, a$ a month 22000

B · University precinct: residents within 1 km 18, workers within 1 km 20, tertiary students within 1 km 40, household income index 80, transit access 0.75, competitors within 500 m 6, rent, a$ a month 12000

C · Suburban rail interchange: residents within 1 km 25, workers within 1 km 12, tertiary students within 1 km 5, household income index 95, transit access 0.7, competitors within 500 m 4, rent, a$ a month 10000

D · Outer-suburb shopping centre: residents within 1 km 20, workers within 1 km 5, tertiary students within 1 km 2, household income index 90, transit access 0.35, competitors within 500 m 2, rent, a$ a month 8000

FEATURES AND WHERE REAL VALUES WOULD COME FROM (NONE RETRIEVED)

Residents within 1 km: stands in for walk-in demand from people living nearby; source to pull: national census, usual residents by small area.

Workers within 1 km: stands in for weekday lunch and after-work demand; source to pull: census place-of-work counts.

Tertiary students within 1 km: stands in for the young, app-first customer base; source to pull: university and TAFE enrolment by campus.

Household income index: stands in for willingness to pay for a premium tea; source to pull: census median household income.

Transit access: stands in for footfall past the counter; source to pull: public-transport timetable data, services per hour at the nearest stops.

Competitors within 500 m: stands in for rival bubble-tea and cafe supply (lowers the score); source to pull: business listings or a map-places count.

Rent, A$ a month: stands in for local cost (lowers the score); source to pull: retail lease listings or an agent quote.

Left out: chagee app searches nearby (needs Chagee app users in the market); delivery orders from the area (needs Chagee on the local delivery platforms); distance to the nearest chagee outlet (there is none yet); sales of existing chagee outlets nearby (there are none yet). They need existing Chagee activity. Missing is unknown, not zero: they are left out, never set to zero, so no area is marked down for a gap every area shares.

INPUTS (ASSUMPTIONS.JSON NEW_MARKET)

weights: residents k 1, workers k 1, students k 1, income index 0.5, transit index 1, competitors 0.5, rent month 0.5 relative weight per feature in the suitability score (0-2) (range 0, 2). Our judgement, illustrative. The new-market ranking slide lets you move each; the sensitivity check re-ranks with each weight set to zero and doubled.

sensitivity: low 0, high 2 multipliers applied to one weight at a time. One-at-a-time sensitivity: each weight times zero (dropped) and times two; the rank range is reported per area.

demand scenarios: store.cups per store per year range and value / store.open days per year cups a day: low, base, high. Placeholder scale only: the Singapore per-outlet volume assumption (itself an assumption, not Chagee data) reused for every Australian area. The score does not set cups: without a calibration from real outlets, every area faces the same range.

forecast error: -0.3, 0, 0.3 share by which the pre-opening forecast misses realised cups a day (range -0.5, 0.5). Scenario assumption: before opening, the forecast cannot know actual demand. Under-forecast, exact, over-forecast.

day mix: moderate 104, large 12 busy and event days a year; the rest of store.open_days_per_year are normal days. Scenario assumption for the Australian pilot: busy days = weekend days; event days = a guess at local public holidays and events, not counted. Weights the year-average contribution on the new-market setup slide.

wage per hour: 36 AUD per hour worked, fully loaded (range 30, 45). Scenario assumption, not sourced. Replace with the applicable award rate plus on-costs before use.

machine cost per unit per month: 2000 AUD per station machine unit per month (range 700, 4000). Scenario assumption, not sourced; the Singapore figure is also an assumption and no exchange rate is applied.

price per cup: 8.5 AUD per cup, in store (range 7, 10). Scenario assumption, not sourced. Check menu prices in the pilot city.

utilities per month: 3000 AUD per outlet per month (range 2000, 4500). Scenario assumption, not sourced.

setups: all hand, heading storyboard.split_presets keys compared. Human-only against machine-made drink, as on slide 11; same station speeds and staffing rule (sim/counter.js).

cannibalisation: overlap km 5, max share 0.3 for a second opening: share of a new outlet's cups taken from the pilot, falling linearly to zero at overlap_km. Scenario assumption. Zero for the first opening: no Chagee outlet is in any catchment.

trigger weeks: 4 consecutive weeks a pilot signal must hold before acting. Our judgement, illustrative.

currency: AUD . Every money figure on the new-market slides and the new-market pilot slide is in Australian dollars (A$). No exchange rate is applied to any Singapore figure.

Reused from the Singapore sections, labelled where used: gross margin per cup (storyboard), the per-outlet volume range (store) as the placeholder demand scale, open hours and days, the hourly demand shape, day-type multipliers, station seconds, staffing rule and minimum crews (sim/counter.js, unchanged).

METHOD (SIM/NEWMARKET.JS)

Score: each feature min-max scaled across the areas (turned round for competitors and rent), weighted mean out of 100. **Sensitivity**: each weight dropped, then doubled, one at a time. **Counter**: equipment sized for the base scenario; realised demand reaches the peak hour as a multiplier on the base arrivals; without AI, one crew all day, set for a normal day's planned peak; with AI, the forecast roster for the day type, the forecast missing realised demand by the stated error. **Contribution**: served cups x price x margin, less labour, machine units, rent and utilities, a day; the peak-hour walk-away share is applied to the busiest hours, other hours assumed served. **Worst case**: the lowest year-average contribution over demand scenarios and forecast errors, with AI staffing.

CLAIMS WE DID NOT MAKE

No Australian data and no Chagee data are used. The areas are archetypes, not suburbs. The score is not a sales forecast; the supervised site model is not validated for Australia. No figure on the new-market slides feeds the main story or the benefit.

SPEAKER NOTES · NEW MARKET: INPUTS AND METHOD

Soft copy only. Backup for the new-market slides: every new-market input, how the numbers are worked out, and what we do not claim.